AI Driven Privacy Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain

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INDIVIDUAL THESIS

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ABSTRACT

Paddy price fluctuations create significant uncertainty for farmers and other stakeholders in Sri Lanka, particularly when making decisions related to harvesting, storage, and marketing. This research component presents an AI driven paddy price forecasting and explainability system designed to provide future paddy price estimates together with understandable evidence behind each prediction. The proposed system utilizes historical paddy price data from multiple Sri Lankan districts and applies an XGBoost based machine learning model to forecast average paddy prices. A recursive multi week forecasting mechanism enables the system to generate sequential future price estimates based on previously generated predictions and historical price patterns. To improve model transparency, SHAP (SHapley Additive exPlanations) is incorporated to identify the features that contribute most significantly to individual predictions and to generate evidence-based explanations, including price trends, confidence levels, market outlook, risk levels, and key influencing factors. Furthermore, an LLM based explanation layer converts the model and SHAP evidence into concise, human friendly explanations without modifying the underlying prediction or introducing unsupported information. The system supports both English and Sinhala, with simplified Sinhala explanations intended to improve accessibility for local farmers. The component is implemented as a FastAPI based backend and exposes prediction, forecasting, and explanation services through RESTful APIs. The resulting system combines predictive analytics, explainable AI, and natural language generation to provide transparent and accessible paddy price intelligence for supporting informed agricultural decision-making in Sri Lanka.

Keywords: Paddy Price Forecasting, Machine Learning, XGBoost, Explainable Artificial Intelligence (XAI), SHAP

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
FastAPI	Fast Application Programming Interface
LLM	Large Language Model
LKR	Sri Lankan Rupee
ML	Machine Learning
NFR	Non-Functional Requirement
REST	Representational State Transfer
SHAP	SHapley Additive exPlanations
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall project context

Sri Lanka's paddy sector plays a significant role in the country's agricultural economy and food supply system. However, the modernization of the paddy supply chain requires more intelligent, secure, and privacy preserving digital technologies capable of supporting farmers and other stakeholders across different stages of the agricultural and market ecosystem. Existing technological approaches generally address individual aspects of the paddy supply chain, such as crop monitoring, yield prediction, price forecasting, traceability, or online marketplaces, without providing an integrated ecosystem that combines intelligent decision-making, privacy protection, secure transactions, and stakeholder interaction.

The proposed research, "AI Driven Privacy Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain," addresses this gap through an integrated ecosystem consisting of four complementary components. The first component focuses on developing a Digital Dashboard for paddy farming, combining real-time data, deep learning, Vision Transformers, Retrieval-Augmented Generation (RAG), and Graph Neural Networks (GNNs) to support yield prediction, crop image analysis, and knowledge-augmented agricultural advisory services. The research specifically identifies the lack of systems that integrate multimodal agricultural inputs such as images, soil, weather, and farm metadata, as well as the limited application of Digital Dashboard, RAG, and GNNs within the Sri Lankan paddy farming context.

The second component addresses the uncertainty associated with paddy price forecasting**.** Conventional statistical and machine learning approaches can predict prices, but they provide limited understanding of the underlying causes of price changes. Therefore, the proposed component combines Transformer based time series forecasting with Causal Analysis Learning, Large Language Model reasoning**.**

The third component focuses on establishing privacy preserving and trustworthy supply chain traceability. Although blockchain can provide tamper resistant records for tracking agricultural products, conventional blockchain based approaches may expose sensitive information about farmers and other participants. To address this limitation, the proposed ecosystem incorporates Zero Knowledge Proofs (ZKPs), Decentralized Identity (DID), smart contracts, and AI based fraud detection using Graph Machine Learning. These technologies are intended to enable stakeholders to verify relevant agricultural and transaction information without unnecessarily exposing sensitive data, while also supporting secure identities and the detection of fraudulent or collusive behaviour within the supply chain.

The fourth component introduces an AI powered farmer miller marketplace designed to improve market participation and decision making while preserving stakeholder privacy. Unlike conventional agricultural marketplaces that primarily display prices, the proposed approach incorporates RAG based LLM assistants, Federated Learning, secure matching, and multi agent negotiation. Federated Learning enables AI models to benefit from distributed farmer data without requiring the underlying private data to be centrally shared. Secure matching mechanisms can further protect sensitive information during buyer seller interactions, while AI agents can support automated negotiation between farmers and millers.

Collectively, these four components establish a unified AI driven, privacy preserving digital ecosystem for Sri Lanka's paddy supply chain. The ecosystem aims to connect farm level intelligence, market forecasting, secure and privacy preserving traceability, and intelligent farmer miller interactions within a common technological framework. By integrating Artificial Intelligence, Digital Dashboard, XGBoost based forecasting, Causal Anaysis Learning, RAG, Graph Neural Networks, Blockchain, Zero Knowledge Proofs, Decentralized Identity, Federated Learning, secure matching, and multi agent systems, the research seeks to move beyond isolated agricultural applications toward a more comprehensive digital ecosystem capable of supporting informed decision making, trustworthy transactions, privacy protection, and efficient stakeholder interaction across the paddy supply chain.

1.2 Individual component context

This research component focuses on developing an AI driven Sri Lankan paddy price forecasting and explainability system. The system uses an XGBoost machine learning model to predict paddy prices for different districts and supports recursive multi-week forecasting. To improve the transparency of the predictions, SHAP based Explainable AI is used to identify important features and their contributions to individual predictions. Additional XAI modules classify market trends, estimate application-level confidence, assess risk, generate market outlooks, and provide decision-support recommendations. An LLM based explanation layer then converts the structured prediction and XAI evidence into human-readable explanations in English and simplified Sinhala, without modifying the underlying machine learning prediction. The component is implemented as a FastAPI based REST backend and provides prediction, forecasting, evidence, and explanation services for integration with the Digital Goviya mobile application. The system has been evaluated using unit,

integration, system, non-functional, and bug fix/regression testing to assess functional correctness, performance, reliability, and robustness.

1.3 Problem statement

Paddy is one of the most important agricultural crops in Sri Lanka and plays a significant role in food security and the livelihoods of farming communities. However, fluctuations in paddy prices create uncertainty for farmers when making decisions related to harvesting, storage, and selling. Prices can vary over time and across districts due to changes in market conditions and other influencing factors. As a result, farmers may find it difficult to determine suitable selling strategies based only on historical observations or current market prices.

Although machine learning techniques can be used to identify patterns in historical agricultural data and predict future paddy prices, conventional prediction systems often provide only a numerical prediction without explaining why a particular price was predicted. This creates a lack of transparency and interpretability, particularly for users who may not have technical knowledge of machine learning. A prediction without supporting evidence may therefore be difficult for farmers and other agricultural stakeholders to understand and use confidently for decision making.

Furthermore, technical machine learning explanations, such as feature contribution values, are not necessarily easy for non-technical users to interpret. There is therefore a need to transform complex model outputs into clear and meaningful explanations while maintaining the reliability of the original prediction. Language accessibility is another important consideration in the Sri Lankan agricultural context, where providing information in Sinhala as well as English can improve accessibility for local users.

Therefore, the problem addressed by this research component is the lack of an integrated paddy price forecasting system that not only predicts future paddy prices but also provides transparent, understandable, and multilingual explanations of the

predictions**.** This component addresses the problem by developing an XGBoost-based paddy price forecasting system supported by SHAP based Explainable AI techniques and an LLM powered explanation layer. The system generates price predictions and multi-week forecasts, identifies important factors contributing to individual predictions, interprets market trends and confidence, and converts the resulting evidence into human-readable English and simplified Sinhala explanations without modifying the underlying machine learning prediction.

1.4 Research gap

Existing research on agricultural price forecasting has demonstrated the potential of machine learning techniques to identify patterns in historical market data and predict future agricultural prices. However, several gaps remain when these approaches are considered in the context of Sri Lankan paddy price forecasting and practical farmer-oriented decision support.

Limited Explainability of Price Forecasting Models

Many machine learning-based forecasting approaches primarily focus on improving prediction performance while providing limited information about why a particular prediction was generated. For paddy price forecasting, a numerical prediction alone may not be sufficient for farmers or other agricultural stakeholders to understand the factors influencing the expected price. This creates a gap between model prediction and practical decision-making.

Lack of Integrated XAI for Paddy Price Prediction

Although Explainable AI techniques such as SHAP can be used to identify the contribution of individual features to machine learning predictions, their integration into a complete paddy price forecasting and decision-support workflow remains limited. There is a need for a system that connects the forecasting model with feature-level explanations, trend interpretation, confidence assessment, risk indication, and recommendations rather than presenting raw model outputs alone.

Limited Farmer-Friendly Interpretation of Technical Explanations

SHAP values and feature contributions are technically meaningful but can be difficult for non-technical users to understand. Existing prediction systems may provide model-level explanations without converting them into simple, actionable language. Therefore, a gap exists in transforming technical XAI evidence into explanations that can be easily understood by agricultural users.

Lack of Multilingual Explanation

Sri Lanka has a multilingual agricultural user base, yet machine learning forecasting systems do not necessarily provide explanations in a form that is accessible to local users. In particular, there is a need to provide simple Sinhala explanations alongside English, rather than limiting the interpretation of AI predictions to technical or English-only outputs.

Limited Integration of LLMs with Evidence-Based XAI

Large Language Models provide an opportunity to convert structured AI outputs into natural-language explanations. However, simply using an LLM to generate explanations can introduce the risk of producing information that is not directly supported by the prediction model. A research gap therefore exists in developing an approach where the LLM is constrained to interpret structured forecasting and XAI evidence rather than independently generating or modifying the prediction.

1.5 Aim and objectives

The aim of this research component is:

To develop an AI-driven, explainable, and multilingual paddy price forecasting system for Sri Lanka that predicts future paddy prices and provides transparent, farmer-friendly explanations to support informed agricultural decision-making.

The proposed approach combines XGBoost-based machine learning, SHAP-based Explainable Artificial Intelligence (XAI), and an LLM-based natural-language explanation layer to improve both the predictive capability and interpretability of paddy price forecasts.

Research Objectives

Objective 1 – Develop a machine learning-based paddy price prediction model.

To develop an XGBoost-based machine learning model capable of predicting paddy prices using historical paddy price data and relevant features for selected districts in Sri Lanka.

Objective 2 – Implement multi-week paddy price forecasting.

To develop a recursive forecasting mechanism that generates weekly paddy price forecasts for a user-defined future period, enabling users to observe expected price movements over multiple weeks.

Objective 3 – Integrate Explainable AI into the forecasting process.

To integrate SHAP-based explainability techniques to identify and quantify the contribution of important features influencing individual paddy price predictions, thereby improving the transparency of the machine learning model.

Objective 4 – Develop meaningful market interpretations from prediction evidence.

To interpret prediction and XAI evidence through understandable indicators such as price trend, confidence level, risk level, market outlook, and decision-support recommendations.

Objective 5 – Develop an evidence-grounded LLM explanation mechanism.

To integrate an LLM-based explanation layer that converts structured machine learning and XAI evidence into understandable natural-language explanations without modifying the underlying model prediction or independently generating unsupported prediction information.

Objective 6 – Provide multilingual and farmer-friendly explanations.

To generate explanations in both English and simplified Sinhala, improving the accessibility of paddy price forecasting information for Sri Lankan farmers and other agricultural stakeholders.

Objective 7 – Develop an API-based forecasting and explanation service.

To implement the proposed forecasting and explainability functions as a FastAPI-based REST service that can be integrated with the Digital Goviya mobile application and other components of the overall digital paddy supply-chain ecosystem.

Objective 8 – Evaluate the developed component.

To evaluate the functionality, integration, performance, reliability, and robustness of the developed system through unit testing, integration testing, system testing, non-functional testing, and regression testing.

1.6 Research questions or hypotheses

For your component, I recommend using research questions rather than formal hypotheses, because your work is primarily focused on developing and evaluating a software-based forecasting and explainability system, rather than testing a controlled experimental treatment.

RQ1. How effectively can an XGBoost-based machine learning model forecast paddy prices for selected districts in Sri Lanka using historical paddy price and relevant feature data?

RQ2. How effectively can recursive multi-week forecasting be used to predict future paddy price movements over multiple weeks?

RQ3. How can SHAP-based Explainable AI be used to identify and communicate the factors contributing to individual paddy price predictions?

RQ4. How can structured machine learning and XAI evidence be transformed into understandable, farmer-oriented explanations using a Large Language Model?

RQ5. To what extent can English and simplified Sinhala explanations improve the accessibility and interpretability of paddy price forecasting information for agricultural stakeholders?

RQ6. How effectively can the proposed forecasting and explainability system provide reliable and responsive API-based services for integration with the Digital Goviya application?

1.7 Scope limitations and assumptions

This research component focuses on developing an AI-driven paddy price forecasting and explainability system for Sri Lanka. The scope covers the development of the machine learning model, multi-week forecasting mechanism, Explainable AI layer,

LLM-based explanation mechanism, and API-based integration required to provide the forecasting functionality to the Digital Goviya ecosystem.

The scope includes the following areas:

Paddy Price Forecasting

The system uses an XGBoost-based machine learning model to predict paddy prices using historical data and relevant available features. The system supports predictions for selected Sri Lankan districts represented in the available dataset.

Multi-Week Forecasting

The component supports recursive weekly forecasting, where a predicted price is incorporated into the forecasting process to generate subsequent future predictions. Users can specify the number of future weeks to forecast.

Explainable AI

The system incorporates SHAP-based Explainable AI to identify important features contributing to individual predictions. The explanation layer further interprets these results through:

- Feature contributions
- Price trend
- Confidence level
- Risk level
- Market outlook
- Recommendations
- Prediction summary

LLM-Based Explanation

An LLM is used to transform structured prediction and XAI evidence into natural-language explanations. The LLM operates as an interpretation layer and does not replace or modify the XGBoost prediction.

Multilingual Support

The explanation mechanism provides outputs in:

- English
- Sinhala

The Sinhala explanations are intended to be understandable to agricultural users rather than being highly technical translations.

API-Based Integration

The forecasting and explanation functionality is implemented using FastAPI REST APIs, allowing the component to communicate with the Digital Goviya mobile application and other components of the overall research system.

System Evaluation

The component is evaluated through:

- Unit testing
- Integration testing
- System testing
- Non-functional testing
- Bug fixing and regression testing

Performance and reliability are evaluated using measured API response times and repeated requests.

Scope Limitations

Despite covering forecasting, explainability, and natural-language interpretation, the component has several limitations.

Dataset Availability

The forecasting performance is dependent on the quality, quantity, and availability of historical data. Missing or incomplete historical information may affect the ability of the system to generate predictions for particular districts or dates.

Limited Feature Availability

The model can only use features available in the prepared dataset. Factors that may influence paddy prices but are not represented adequately in the dataset cannot be directly incorporated into the prediction.

District Coverage

The system only supports districts represented by sufficient historical information in the available dataset. It does not claim to provide reliable predictions for every geographical location in Sri Lanka.

Forecasting Uncertainty

Future paddy prices are affected by changing market and environmental conditions. Therefore, forecasts should be considered estimates rather than guaranteed future prices.

Recursive Forecasting Error

The multi-week forecasting mechanism uses previous predictions as inputs for subsequent predictions. Consequently, prediction errors may accumulate as the forecasting horizon increases.

XAI Limitations

SHAP identifies feature contributions to model predictions, but these contributions should not automatically be interpreted as proof of causal relationships. A feature contributing strongly to a prediction does not necessarily mean that it directly caused the price movement.

Confidence Interpretation

The confidence classification provided by the system is an application-specific confidence assessment based on available model explanation evidence. It is not a statistically calibrated probability that the prediction will be correct.

LLM Limitations

The LLM is used to explain structured evidence generated by the forecasting and XAI components. Although the explanation mechanism is designed to remain grounded in this evidence, natural-language generation can still have limitations associated with language-model interpretation. The LLM should therefore not be considered an independent source of agricultural or market truth.

Recommendation Limitations

The recommendations generated by the system are intended as decision-support information. They do not constitute guaranteed financial, commercial, or agricultural advice.

Performance Testing Limitation

The repeated-request scalability evaluation was conducted using sequential requests. Therefore, the measured results do not demonstrate the performance of the system under a large number of simultaneous or concurrent users.

Language Limitation

The multilingual explanation component is limited to English and simplified Sinhala. Other Sri Lankan languages are outside the scope of this component.

External Market Factors

The system does not guarantee that every sudden external event affecting paddy prices can be captured by the forecasting model. Unexpected changes in market conditions, agricultural policies, weather, supply, demand, or other external factors may reduce forecasting accuracy.

Assumptions

The development and evaluation of the component are based on the following assumptions.

Data Assumption

It is assumed that the historical paddy price dataset used for model development is sufficiently representative of the patterns required for forecasting within the selected districts and forecasting period.

Data Quality Assumption

It is assumed that the prepared and cleaned dataset contains sufficiently accurate values for the features used by the XGBoost model.

Feature Consistency Assumption

It is assumed that the input features provided during prediction are consistent with the feature structure and preprocessing used during model development.

Historical Pattern Assumption

The forecasting approach assumes that patterns observed in historical data provide useful information about future paddy price movements.

Model Assumption

It is assumed that the trained XGBoost model remains valid for the data distribution encountered during the prediction period. Significant changes in the underlying market conditions may require future model retraining.

SHAP Assumption

It is assumed that SHAP provides useful evidence for interpreting the relationship between the model's input features and individual predictions, while recognizing that SHAP explanations represent model behavior rather than causal relationships.

LLM Grounding Assumption

It is assumed that providing structured prediction and XAI evidence to the LLM allows the generated explanations to remain consistent with the underlying model outputs.

API Availability Assumption

It is assumed that the FastAPI backend and required model/data resources are available when the Digital Goviya application requests forecasting or explanation services.

User Interpretation Assumption

It is assumed that farmers and other intended users can understand the simplified Sinhala and English explanations sufficiently to use the information as supplementary decision-support information.

Forecast Horizon Assumption

It is assumed that shorter forecasting horizons provide more meaningful predictions than very long recursive forecasting horizons because prediction uncertainty may increase as additional future predictions are generated.

1.8 Individual contribution

As an individual researcher and developer, my contribution to the overall research project focused on the design, development, integration, and evaluation of the AI-driven paddy price forecasting and explainability component. The main contributions are described below.

Machine Learning-Based Paddy Price Forecasting

I developed the machine learning component for forecasting paddy prices using an XGBoost-based prediction model. This included preparing the required data and implementing the prediction workflow to generate paddy price predictions for supported districts.

The component was designed to accept a district and prediction date and return the corresponding predicted paddy price through a REST API.

Multi-Week Forecasting

I implemented a recursive multi-week forecasting mechanism that generates future weekly paddy price predictions.

The forecasting process uses each generated prediction as part of the subsequent forecasting process, allowing the system to generate a user-defined number of future weekly predictions.

Explainable AI Implementation

A major part of my individual contribution was the development of the Explainable AI (XAI) layer using SHAP.

I implemented separate modules for:

- Trend classification
- Confidence estimation
- Feature importance
- Market outlook
- Risk assessment
- Recommendations
- Prediction summaries

This enables the system to provide not only a predicted price but also information explaining the model's prediction.

SHAP-Based Feature Explanation

I integrated SHAP-based explanations to identify the features that contribute most significantly to individual predictions.

The system extracts the important feature contributions and converts them into human-readable reasons. This provides evidence that can be used by the subsequent explanation layer rather than relying only on the numerical prediction.

LLM-Based Explanation Layer

I designed and implemented an LLM-powered explanation layer that converts structured machine learning and XAI evidence into natural-language explanations.

The LLM receives information such as:

- Predicted price
- Previous price
- Market trend
- Confidence
- Risk level
- Market outlook
- Recommendation
- Important features
- SHAP-based reasons

The LLM is used only as an interpretation layer and does not modify the original XGBoost prediction.

Multilingual Explanation

I implemented support for English and simplified Sinhala explanations.

The Sinhala explanation was designed to use simpler terminology so that the generated information is more accessible to non-technical agricultural users. The response structure was maintained consistently for both languages.

FastAPI Backend Development

I developed the backend service using FastAPI and exposed the forecasting and explanation functionality through REST API endpoints.

The main endpoints developed include:

POST /predict

POST /forecast

POST /explanation/evidence

POST /explanation/llm

POST /prediction/explanation

These APIs allow the component to be integrated with the Digital Goviya mobile application and other components of the overall research system.

Backend Service Architecture

I organized the backend into separate service layers to improve maintainability and separation of responsibilities.

The implementation includes services for:

- Data loading
- Model loading
- Feature generation
- Prediction
- Forecasting
- XAI explanation
- SHAP explainer management
- LLM explanation
- LLM communication

This structure separates machine learning operations from API handling and natural-language explanation.

Error Handling and Validation

I implemented centralized exception handling and input validation for the forecasting APIs.

As part of this work, I identified and fixed a defect where an unsupported district was incorrectly reported as having insufficient historical data.

The solution added district validation before historical feature generation, resulting in a clearer error response:

“The selected district is not supported.”

Testing and Evaluation

I was responsible for testing and evaluating the developed component through multiple testing levels:

- Unit testing for XAI modules
- Integration testing for API and service interactions
- System testing for complete API workflows
- Non-functional testing for performance and reliability
- Bug fixing and regression testing

Performance measurements were also collected for the major APIs, including /predict, /forecast, and /explanation/llm.

1.9 Thesis structure

Chapter 1 – Introduction

This chapter introduces the research component and establishes the background and motivation for developing an AI-driven paddy price forecasting and explainability system for Sri Lanka. It presents the problem addressed by the research, research gap, aim, objectives, research questions, scope, limitations, assumptions, and individual contribution.

The chapter provides the overall context for the subsequent chapters and explains the importance of combining machine learning-based forecasting with Explainable AI and natural-language explanations.

Chapter 2 – Literature Review

This chapter reviews existing research and technologies relevant to the proposed system. It discusses:

- Paddy and agricultural price forecasting
- Agricultural price prediction using machine learning
- XGBoost and tree-based forecasting approaches
- Time-series and recursive forecasting
- Explainable Artificial Intelligence
- SHAP-based model explanation
- Large Language Models
- LLM-based explanation systems
- Multilingual and farmer-oriented AI systems
- Existing agricultural decision-support systems

The chapter also compares existing approaches and identifies the research gap addressed by the proposed component.

Chapter 3 – Methodology

This chapter presents the methodology followed to design and develop the proposed system. It describes the overall research and development process, including:

- Research approach
- Data collection and preparation
- Data preprocessing
- Feature engineering
- XGBoost model development
- Forecasting methodology
- SHAP-based explainability
- XAI interpretation
- LLM explanation methodology
- Multilingual explanation generation
- FastAPI service development
- System integration approach
- Evaluation methodology

The chapter explains how the different stages are connected to achieve the research objectives.

Chapter 4 – System Analysis and Design

This chapter presents the analysis and design of the proposed forecasting and explainability system.

It includes:

- Functional requirements
- Non-functional requirements
- System architecture
- Component architecture
- Data flow
- API architecture
- Service-layer design
- Machine learning workflow
- XAI workflow
- LLM explanation workflow
- Database/data structure where applicable
- API request and response structures
- Error-handling design

The relationship between the React Native application, FastAPI backend, XGBoost model, SHAP explanation layer, XAI modules, and LLM explanation layer is also presented.

Chapter 5 – Implementation

This chapter describes the implementation of the proposed system.

It covers the development of:

- Data loading and processing
- Feature generation
- XGBoost prediction service
- Recursive multi-week forecasting

- SHAP explainer
- Trend classification
- Confidence estimation
- Feature importance analysis
- Market outlook
- Risk assessment
- Recommendation generation
- Prediction summary
- LLM explanation service
- English and simplified Sinhala explanations
- FastAPI REST APIs
- Validation and exception handling

Relevant implementation details and selected code/design decisions are discussed in this chapter.

Chapter 6 – Testing and Evaluation

This chapter evaluates the developed system against the defined objectives and requirements.

The evaluation includes:

Functional Testing

- Prediction testing
- Forecasting testing
- XAI evidence testing
- LLM explanation testing
- Input validation
- Error handling

Unit Testing

Testing of individual XAI and supporting modules.

Integration Testing

Testing interactions between:

API → Services → Model → XAI → LLM

System Testing

Testing complete user-request workflows through the API.

Non-Functional Testing

Including:

- API response-time measurements
- Reliability testing
- Repeated-request performance testing

Bug Fixing and Regression Testing

Documenting identified defects, root causes, implemented fixes, and subsequent regression testing.

The chapter presents the measured results and discusses whether the developed system satisfies the defined requirements and research objectives.

Chapter 7 – Results and Discussion

This chapter presents and discusses the results obtained from the developed system.

It discusses:

- Paddy price prediction results
- Multi-week forecasting results
- SHAP explanation results
- Trend and confidence results
- Market outlook and recommendations
- LLM-generated explanations
- English and Sinhala outputs
- API performance results
- Reliability results
- Testing outcomes

The results are interpreted in relation to the research questions and objectives. The strengths and weaknesses of the developed approach are also discussed.

Chapter 8 – Conclusion and Future Work

This chapter concludes the research component by summarizing the work carried out and the major findings.

It includes:

- Summary of the developed system
- Achievement of research objectives
- Answers to the research questions
- Main contributions
- Research significance
- Limitations identified during development and evaluation
- Recommendations for future improvements

Potential future work may include:

- Expanding the dataset and district coverage
- Incorporating additional market and environmental variables
- Improving long-horizon forecasting
- Model retraining with continuously updated data
- More extensive concurrent load testing
- Supporting additional Sri Lankan languages
- Further evaluation with real agricultural users

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review for this research component focuses on identifying and examining existing work related to paddy price forecasting, machine learning-based agricultural price prediction, Explainable AI (XAI), SHAP-based model interpretation, and Large Language Model (LLM)-based explanations.

The review is structured around the main technical areas of the proposed system:

Agricultural and Paddy Price Forecasting

Existing approaches for predicting agricultural and paddy prices are reviewed to understand the commonly used data sources, forecasting methods, machine learning techniques, and challenges associated with agricultural price prediction.

Machine Learning for Price Prediction

Machine learning approaches, particularly tree-based models and XGBoost, are reviewed to understand their suitability for modelling nonlinear relationships within historical agricultural price data.

Explainable Artificial Intelligence

Research on XAI is reviewed to identify methods that can explain machine learning predictions and improve model transparency. Particular attention is given to SHAP, as it provides feature-level contribution information for individual predictions.

Natural-Language and LLM-Based Explanations

Existing research involving Large Language Models for interpreting or communicating machine learning outputs is reviewed. The review considers how structured model evidence can be transformed into understandable natural-language explanations while maintaining consistency with the underlying prediction.

Multilingual and User-Oriented AI Systems

Research related to accessible and multilingual AI-based decision-support systems is considered, particularly approaches that can communicate technical information to non-technical users.

Identification of the Research Gap

Findings from the reviewed studies are compared with the requirements of the proposed component to identify limitations in existing approaches and establish the need for an integrated system combining paddy price forecasting, XAI, and evidence-based LLM explanations.

Review Scope

The review is limited to literature and existing approaches directly relevant to the development of the proposed paddy price forecasting and explainability component.

The review covers:

- Paddy and agricultural commodity price forecasting
- Agricultural price prediction using machine learning
- XGBoost and related machine learning approaches
- Time-series and multi-step forecasting
- Explainable Artificial Intelligence
- SHAP and feature-level model explanations
- LLM-based explanation and interpretation
- Evidence-grounded natural-language generation
- Multilingual AI explanations
- Agricultural decision-support systems

The review primarily focuses on studies that can inform the technical design, explainability, usability, and evaluation of the proposed system.

The review does not attempt to provide a comprehensive review of all agricultural technologies, the complete Sri Lankan paddy supply chain, agricultural IoT systems, blockchain implementations, or unrelated AI applications. These areas belong to other components or broader aspects of the overall research project and are outside the specific scope of this component.

Review Selection Focus

The reviewed literature is assessed based on factors such as:

- Relevance to agricultural or paddy price forecasting

- Machine learning techniques used
- Forecasting methodology
- Availability of model explanations
- Use of SHAP or other XAI techniques
- Use of LLMs for explanation
- Support for non-technical users
- Multilingual or local-language accessibility
- Applicability to the Sri Lankan agricultural context

The outcome of this review provides the theoretical and technical foundation for selecting XGBoost for forecasting, SHAP for model explainability, and an evidence-grounded LLM for human-readable English and simplified Sinhala explanations in the proposed system.

2.2 Domain and theoretical foundations

This section establishes the theoretical foundation of the proposed Sri Lankan paddy price forecasting and explainability component. The component combines concepts from agricultural price forecasting, machine learning, Explainable Artificial Intelligence (XAI), SHAP-based interpretation, recursive forecasting, and Large Language Models (LLMs). These concepts provide the theoretical basis for the design and implementation of the proposed system.

Agricultural Price Forecasting

Agricultural commodity prices are influenced by multiple interacting factors and can change over time. For farmers and other agricultural stakeholders, anticipating future price movements can support decisions related to harvesting, storage, and marketing.

In the context of this research, paddy price forecasting refers to estimating future paddy prices using historical observations and relevant features available within the dataset. Rather than relying only on the latest observed price, the proposed approach uses a machine learning model to identify patterns in historical data and generate future price estimates.

The forecasting component is designed to provide both single-date predictions and multi-week forecasts. This makes the system suitable for integration into a digital agricultural decision-support environment.

Concept	Description	Application in This Component
Historical Price	Previously observed paddy price information	Used as input for model learning and forecasting
Prediction	Estimated paddy price for a specified date	Generated using the XGBoost model
Forecast Horizon	Number of future periods to be predicted	User-defined number of weeks
District	Geographic area associated with price observations	Used to generate district-specific predictions
Recursive Forecasting	Sequential generation of future predictions	Used for multi-week forecasting

Table 1 : Key Concepts in Paddy Price Forecasting

Machine Learning for Price Prediction

Machine learning provides methods for learning relationships and patterns from historical data and applying those learned relationships to previously unseen inputs.

For this research component, machine learning is used as the core predictive mechanism. The model receives the required features for a particular district and date and produces an estimated paddy price.

The use of machine learning is particularly relevant because agricultural price relationships may be nonlinear and may involve interactions between multiple variables. A tree-based ensemble model can capture such relationships without requiring a simple linear relationship between the input variables and the predicted price.

XGBoost

Extreme Gradient Boosting (XGBoost) is a gradient-boosting-based ensemble learning algorithm that builds a collection of decision trees sequentially. Each subsequent tree contributes to improving the prediction by addressing errors made by previous trees.

Conceptual XGBoost Prediction Process

The proposed component uses an XGBoost model because the forecasting task involves structured tabular data and potentially nonlinear relationships between features and paddy prices.

Multi-Step Recursive Forecasting

The proposed component supports forecasting beyond a single future date through recursive multi-step forecasting.

In recursive forecasting, the prediction for one future period can be incorporated into the input history used to generate the next prediction.

Recursive Multi-Week Forecasting Workflow

For example, if six weeks are requested, the system generates the first week's prediction and subsequently uses the updated historical sequence to generate predictions for the following weeks.

A limitation of this approach is that errors can propagate through later forecasting steps because previous predictions become part of the subsequent forecasting process.

Explainable Artificial Intelligence

Machine learning models can produce accurate predictions while making it difficult for users to understand the reasoning behind individual predictions. Explainable Artificial Intelligence (XAI) addresses this issue by providing information about how a model arrives at its outputs.

In this research, XAI is important because the system is intended not only to predict paddy prices but also to provide evidence that helps users understand the prediction.

SHAP-Based Explanation

SHAP (SHapley Additive exPlanations) is used in the proposed system to identify the contribution of individual features to a particular model prediction.

For a given prediction, SHAP provides contribution values that indicate how individual features influence the model output relative to the model's baseline.

Conceptual SHAP Feature Contribution Process

In the proposed component, SHAP information is further processed by the feature_importance.py module to identify important contributing features and generate human-readable reasons.

Large Language Models

Large Language Models are capable of generating natural-language text from structured or unstructured information. In the proposed system, the LLM is not used as the primary forecasting model.

LLM-Based Explanation Architecture

This separation is important because the numerical prediction remains controlled by the machine learning model, while the LLM is responsible for communicating the available evidence in a form that is easier for users to understand.

Evidence-Grounded LLM Explanation

A key design principle of the proposed system is that the LLM should explain existing evidence rather than independently determine the paddy price.

The structured evidence provided to the LLM includes information such as:

- District
- Prediction date
- Predicted price
- Previous price
- Trend
- Confidence
- Risk level

- Market outlook
- Recommendation
- Top contributing features
- SHAP-based reasons

Multilingual and Farmer-Oriented AI

A technically correct prediction may have limited practical value if the intended users cannot easily understand its meaning. Therefore, the proposed system provides explanations in English and simplified Sinhala.

The multilingual explanation layer is designed to communicate:

- What the predicted price is
- Whether the expected trend is increasing, decreasing, or stable
- Which factors influenced the prediction
- What the model evidence indicates
- What general decision-support information can be considered

The purpose is not to replace technical XAI evidence but to make the evidence more accessible to users who may not have a technical background in machine learning.

Theoretical Foundation of the Proposed Component

The theoretical foundations of the component can be summarized as a layered approach:

Layer	**Theoretical Concept**	**Technology/Approach**	**Purpose**
Prediction	Machine Learning	XGBoost	Predict paddy prices
Forecasting	Recursive Multi-Step Forecasting	Sequential prediction	Generate future weekly prices

Explainability	Explainable AI	SHAP	Identify feature contributions
Interpretation	Model Evidence Interpretation	XAI modules	Generate trend, confidence, risk and outlook
Communication	Natural Language Generation	LLM	Convert evidence into understandable explanations
Accessibility	Multilingual AI	English + simplified Sinhala	Improve user accessibility
Integration	RESTful Service Architecture	FastAPI	Provide forecasting and explanation services

Table 2 : Theoretical Foundations of the Proposed System

2.3 Related research

Related Research

Paddy price forecasting has been investigated using statistical time-series models and machine learning techniques to support agricultural decision-making and reduce the uncertainty associated with price fluctuations. Existing research demonstrates that forecasting models can provide useful estimates of future paddy prices; however, many of the existing approaches primarily focus on predicting future prices from historical price patterns and provide limited insight into the external factors and causal mechanisms responsible for price changes.

Kathayat and Dixit (2021) investigated paddy price forecasting in India using the Autoregressive Integrated Moving Average (ARIMA) model. The study focused on monthly wholesale paddy prices across five major markets in Punjab, Tamil Nadu, Delhi, West Bengal, and Uttar Pradesh. Historical price data covering July 2002 to June 2020 were obtained from the AGMARKNET price portal. The researchers

analysed seasonal price variations and applied ARIMA models to forecast paddy prices for the 2020–21 agricultural year. Different ARIMA configurations were selected for each market, including ARIMA(4,0,12) for Punjab, ARIMA(0,1,6) for Tamil Nadu, ARIMA(0,1,12) for Delhi, ARIMA(0,1,3) for West Bengal, and ARIMA(3,1,12) for Uttar Pradesh. Model performance was evaluated using RMSE and MAPE, while diagnostic checking was performed using the Ljung–Box test.

The study demonstrated that ARIMA models can effectively capture historical and seasonal patterns in paddy prices and can be used to generate short-term forecasts. It also highlighted considerable differences in price behaviour between geographical markets, with seasonal variations being particularly prominent in some regions. However, the methodology is primarily univariate, relying on historical price observations to predict future prices. Although the study discusses factors such as production cycles and government Minimum Support Price mechanisms, these external factors are not integrated as explicit explanatory variables within the forecasting model. Furthermore, the study focuses on conventional statistical forecasting and does not provide mechanisms for explaining the causal factors behind individual price movements.

Similarly, Ramadhani, Sukiyono, and Suryanty (2020) investigated paddy grain and rice price forecasting in Indonesia using ARIMA models. Their study considered price levels across different stages of the market, including the farmer, milling, wholesale, and international levels. The researchers first assessed stationarity and then selected suitable ARIMA configurations based on forecasting criteria including the Akaike Information Criterion (AIC) and Schwarz Criterion (SC). The resulting models were ARIMA(1,1,2) for farmer-level and milling-level paddy grain prices, ARIMA(1,1,3) for wholesale rice prices, and ARIMA(3,1,7) for international rice prices. The study subsequently generated six-month forecasts and compared the predicted values with actual prices.

An important contribution of this study is its recognition that different levels of the rice market may require different forecasting models. The findings showed that a single ARIMA configuration is not necessarily appropriate for all market levels. However, similar to the Indian study, the approach remains primarily dependent on

historical price information and does not incorporate a broader set of economic, climatic, or policy-related variables into the forecasting framework. Consequently, while the model can estimate future prices, it provides limited capability for determining why a particular price movement occurs or how a change in an external factor could influence future prices.

A more machine-learning-oriented approach was presented by Sarkar et al. (2022) in their study *Paddy Price Prediction in the South-Western Region of Bangladesh*. The research investigated paddy price prediction using data collected from farmers in Bangladesh's Khulna Division. The dataset contained 154 instances and ten attributes, including division, district, sub-district, market, pricing type, crop price, crop quantity, date, crop category, and crop name. The researchers compared Linear Regression with a Neural Network approach. The reported results showed an RMSE of 114.48 and MAE of 80.08 for Linear Regression, while the Neural Network produced an RMSE of 338.2241 and MAE of 293.1295. Based on these results, Linear Regression performed better than the Neural Network for the dataset used in the study.

This study demonstrates the potential of supervised machine-learning methods for incorporating multiple attributes into agricultural price prediction rather than relying exclusively on historical price sequences. Nevertheless, the study was conducted using a relatively small dataset collected from only three districts in the Khulna Division. The authors also reported that the application of SMOTE did not provide particular benefits because of the characteristics and limited size of the dataset. More importantly, the approach remains focused on predictive accuracy and does not investigate causal relationships between external factors and paddy prices. The study therefore provides evidence for the application of machine learning to paddy price prediction, while also demonstrating the challenges associated with limited agricultural datasets.

Existing research	**Main approach**	**Main contribution**	**Limitation relevant to your work**

Kathayat & Dixit (2021)	ARIMA	Forecasts monthly paddy prices across Indian markets	Mainly historical/univariate forecasting; limited causal/external-factor analysis
Ramadhani et al. (2020)	ARIMA	Forecasts prices at farmer, milling, wholesale and international levels	Different ARIMA models are required, but causal reasoning is absent
Sarkar et al. (2022)	Linear Regression + Neural Network	Uses multiple attributes for Bangladesh paddy price prediction	Small dataset; predictive focus; no causal reasoning or market simulation

Table 3 : Related Research

2.4 Existing systems products and techniques

The reviewed studies demonstrate a progression from conventional statistical forecasting toward machine-learning-based agricultural price prediction. The Indian study by Kathayat and Dixit demonstrates the application of ARIMA to long-term historical wholesale paddy price data across multiple markets, while Ramadhani et al. extend ARIMA forecasting across different stages of the rice market in Indonesia. Sarkar et al. subsequently demonstrate the application of supervised machine learning through Linear Regression and Neural Networks using multiple agricultural and market attributes.

However, several limitations remain evident across these studies. First, the ARIMA-based approaches primarily model historical price behaviour and therefore have limited capability to incorporate multiple external factors simultaneously. Second, although the Bangladesh study incorporates several attributes, its prediction framework does not provide causal reasoning regarding how individual factors influence price changes. Third, none of the reviewed studies provides an integrated mechanism for combining long-range time-series forecasting with causal analysis and knowledge-based explanation. Finally, the reviewed studies do not simulate the

interactions among different stakeholders in the paddy supply chain, such as farmers, millers, wholesalers, and government actors.

These limitations establish the research gap addressed by the proposed **AI-Driven Paddy Price Forecasting** component. The proposed approach extends conventional forecasting by combining **XGBoost-based models, Causal Analysis, Large Language Model reasoning, and multi-agent simulation**. Causal Analysis is used to investigate the factors contributing to price fluctuations rather than relying solely on correlations. The SHAP reasoning layer provides explanations using relevant economic and policy information.

Therefore, the key distinction of the proposed research is that it does not treat paddy price forecasting solely as a numerical prediction problem. Instead, it aims to develop a more comprehensive framework capable of **forecasting future paddy prices, identifying factors that influence price movements, explaining the reasons behind predicted changes, and simulating potential market behaviour**. This addresses the identified research gap in developing a unified framework for forecasting, explaining, and simulating paddy price changes within the Sri Lankan context.

2.5 Gap synthesis and conceptual frameworks

Gap Synthesis

The review of related research, systems, products, and techniques indicates that significant progress has been made in agricultural price forecasting, machine learning, Explainable Artificial Intelligence (XAI), and Large Language Model (LLM)-based explanation. However, these areas are often investigated independently or with different application objectives.

Machine learning research demonstrates that algorithms such as XGBoost can be applied to agricultural and rice-price forecasting. However, forecasting-focused systems generally emphasize the generation of accurate numerical predictions and provide limited support for explaining individual predictions to non-technical users.

Similarly, XAI research demonstrates that techniques such as SHAP can identify the contribution of individual features to machine learning predictions. However, technical SHAP values and feature contributions may not be directly understandable to farmers or other agricultural stakeholders. Therefore, an additional interpretation layer is required to transform technical evidence into meaningful information.

Research involving LLMs provides an opportunity to convert structured machine learning explanations into natural-language descriptions. However, the LLM should not independently determine the prediction because this could separate the generated explanation from the actual evidence produced by the forecasting model. Therefore, an evidence-grounded approach is more appropriate, where the LLM receives structured prediction and XAI evidence and communicates that evidence to the user.

Another identified gap is local-language accessibility. Although machine learning and XAI systems can provide technically detailed explanations, fewer approaches focus specifically on presenting agricultural forecasting information in simple Sinhala alongside English for Sri Lankan users.

Identified Gap	**Existing Research/Systems**	**Requirement Identified**	**Proposed Solution**
Numerical prediction without sufficient explanation	Forecasting-focused ML systems	Explain individual predictions	XGBoost + SHAP
Technical XAI outputs	SHAP-based systems	Convert technical evidence into understandable information	XAI interpretation layer
Limited integration of forecasting and XAI	Prediction and XAI often treated separately	Unified prediction-explanation workflow	Integrated architecture

Difficulty interpreting SHAP values	Technical explanation approaches	Human-readable explanations	LLM explanation layer
Risk of unsupported generated explanations	General LLM-based generation	Ground explanations in model evidence	Evidence-grounded LLM
Limited local-language accessibility	Many AI systems provide English/technical outputs	Support local users	English + simplified Sinhala
Limited farmer-oriented interpretation	Technical prediction systems	Decision-support information	Trend, confidence, risk, outlook and recommendation
Limited end-to-end implementation	Individual techniques evaluated separately	Integrated deployable service	FastAPI-based backend

Table 4 : Synthesized Research Gaps

Conceptual Framework

The conceptual framework illustrates the relationship between the input data, machine learning forecasting process, Explainable AI, LLM interpretation, and the resulting decision-support outputs.

The proposed framework is based on the principle that the final user-facing explanation should remain connected to the evidence generated by the forecasting model.

Components of the Conceptual Framework

Input Layer

The input layer contains the historical paddy price data and the information required to generate a prediction, including the relevant district and date.

The quality and availability of these inputs influence the ability of the forecasting model to generate predictions.

Processing Layer

The processing layer prepares the required information and generates the features expected by the trained XGBoost model.

This layer provides the connection between the raw historical data and the machine learning prediction process.

Prediction Layer

The XGBoost model produces the predicted paddy price.

The same prediction mechanism supports recursive multi-week forecasting, where each generated prediction can contribute to the subsequent forecasting step.

Explainability Layer

SHAP is applied to the prediction to identify feature-level contributions.

These contributions form the evidence used by the XAI modules to derive:

- Trend
- Confidence
- Risk
- Market outlook
- Recommendation
- Summary

Natural-Language Explanation Layer

The structured evidence is passed to the LLM. The LLM's responsibility is to communicate the existing evidence, rather than independently calculate or modify the predicted price.

Output Layer

The final output provides both technical and user-oriented information. The system can return the predicted price and structured evidence while also providing natural-language explanations in English and simplified Sinhala.

Input–Process–Output Framework

A second, simplified conceptual representation can be used to clearly demonstrate the overall logic of the component.

Stage	Components	Output
Input	Historical paddy data, district, date, model features	Prepared prediction data
Prediction	XGBoost	Paddy price prediction
Forecasting	Recursive forecasting	Multi-week price forecasts
Explanation	SHAP	Feature contributions
Interpretation	XAI modules	Trend, confidence, risk, outlook, recommendation
Communication	Evidence-grounded LLM	Natural-language explanation
Accessibility	English + simplified Sinhala	User-friendly multilingual output
Decision Support	Combined information	Interpretable paddy price intelligence

Table 5 : Input–Process–Output Framework

Alignment Between Research Gap and Proposed Framework

The conceptual framework directly addresses the gaps identified through the literature review.

Research Gap	Framework Component	Expected Contribution

Lack of interpretable price predictions	XGBoost + SHAP	Prediction with feature-level evidence
Technical model explanations	XAI interpretation layer	Simplified interpretation
Lack of integrated forecasting and explanation	End-to-end architecture	Unified workflow
Difficulty understanding SHAP values	LLM explanation	Natural-language interpretation
Potential LLM hallucination	Evidence-grounded LLM	Explanation based on model evidence
Limited local-language accessibility	Sinhala explanation	Improved accessibility
Limited farmer-oriented information	Trend, risk, outlook and recommendation	Decision-support information
Need for application integration	FastAPI REST API	Integration with Digital Goviya

Table 6 : Research Gap to Conceptual Framework Mapping

3 RESEARCH METHODOLOGY

3.1 Research design

This research adopts a design-and-development-oriented research design to develop and evaluate an AI-driven paddy price forecasting and explainability component for the Sri Lankan agricultural context. The research focuses on designing a software-based solution that combines machine learning-based forecasting, Explainable Artificial Intelligence (XAI), and Large Language Model (LLM)-based natural-language explanation.

The research design follows an iterative process in which the problem is first identified and investigated through the literature review, followed by data preparation, model development, explainability implementation, system integration, and empirical evaluation.

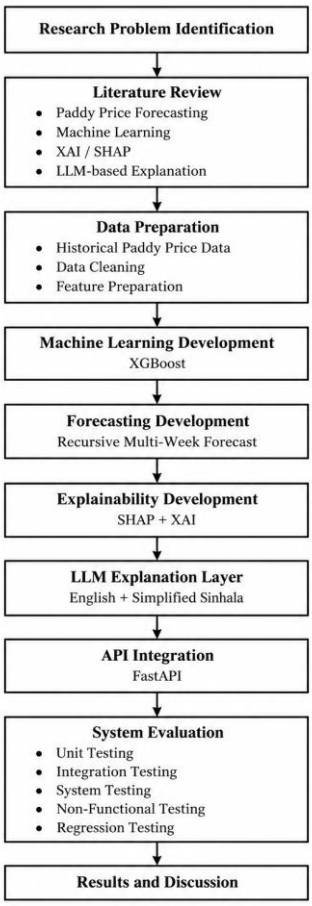


Figure 1 : Research Design and Development Process of the Proposed Component

Research Approach

The research follows a quantitative and experimental evaluation approach for the prediction and system evaluation aspects, combined with a design and development approach for constructing the software component.

The quantitative aspect is reflected in the use of measurable outputs such as:

- Predicted paddy prices
- Multi-week forecast values
- SHAP feature contributions
- API response times
- Request success rates
- Unit and integration test results

The development aspect focuses on constructing an operational system that integrates:

XGBoost → Recursive Forecasting → SHAP → XAI → LLM → FastAPI

The developed component is subsequently evaluated using predefined functional and non-functional tests.

Alignment with Research Objectives

Research Objective	Research Design Activity	Evaluation Method
Develop XGBoost-based paddy price prediction	Model development	Prediction/system testing
Implement multi-week forecasting	Recursive forecasting development	Forecast integration and system testing
Integrate SHAP-based XAI	XAI development	Unit and integration testing
Generate meaningful interpretations	Trend, confidence, risk, outlook and recommendation modules	Unit/system testing

Develop LLM-based explanations	Evidence-grounded LLM integration	LLM integration/system testing
Provide English and simplified Sinhala	Multilingual explanation implementation	Output validation
Provide API-based integration	FastAPI development	API integration testing
Evaluate system quality	Testing and evaluation	NFR and regression testing

Table 7 : Research Design and Objective Alignment

3.2 Data participants or study materials

This research component focuses on developing and evaluating an AI-driven paddy price forecasting and explainability system. Unlike research involving human participants, surveys, or interviews, the primary materials used in this component are historical paddy price data, a trained machine learning model, software frameworks, and computational tools. The study evaluates the developed system through model outputs, API responses, and software testing.

Study Materials

The study materials consist of the data, trained model, software libraries, and development tools required to implement the forecasting and explainability pipeline.

Dataset Description

The primary data material is the prepared historical paddy price dataset, stored in CSV format. It is loaded by the backend and used to generate the features required by the trained XGBoost model.

The dataset supports district-specific prediction and recursive multi-week forecasting. During forecasting, the system uses historical observations to generate features, predicts the next week's price, and incorporates the prediction into the updated data sequence for subsequent forecasting steps.

Machine Learning and Explainability Materials

The component uses a trained XGBoost model to produce price predictions. SHAP is then used to obtain feature-level evidence associated with individual predictions. The evidence is processed by the XAI modules to generate trend classifications, confidence categories, risk levels, market outlooks, and recommendations.

The LLM explanation service receives this structured evidence and generates human-readable explanations in English and simplified Sinhala. The generated explanation is intended to communicate the model evidence rather than independently determine the predicted price.

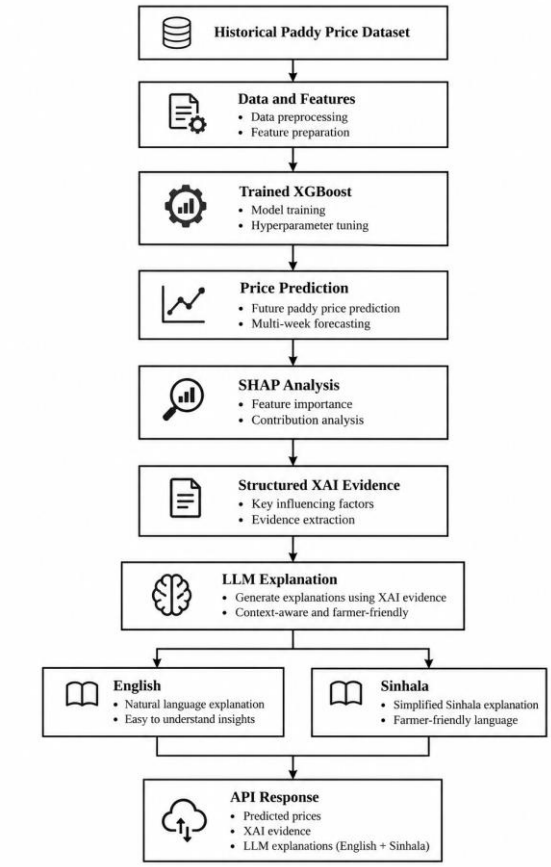


Figure 2 : Study Materials and Processing Pipeline of the Proposed Component

3.3 Experimental or study procedure

The study procedure describes the sequence of activities followed to develop, integrate, and evaluate the proposed AI-driven paddy price forecasting and explainability component. The procedure begins with preparing the historical paddy

price data and configuring the trained XGBoost model, followed by implementing multi-week forecasting, integrating SHAP-based Explainable AI, generating LLM-based explanations, and evaluating the completed backend through software testing.

The study follows a structured development and evaluation process to ensure that each component performs its intended responsibility and that the integrated system produces appropriate outputs for the Digital Goviya application.

3.4 Measures and evaluation metrics

This section describes the measures and evaluation metrics used to assess the performance and functionality of the proposed AI-driven paddy price forecasting and explainability component. The evaluation focuses on two main areas: forecasting and explanation functionality, and software system performance and reliability.

The component combines an XGBoost-based prediction model, recursive multi-week forecasting, SHAP-based explainability, XAI interpretation modules, an LLM explanation service, and FastAPI REST endpoints. Therefore, evaluating the component requires more than measuring prediction accuracy alone. The evaluation also considers API response time, request success rate, correct handling of invalid inputs, integration between services, and regression testing after bug fixes.

The metrics presented below distinguish between measurements already collected during system testing and additional metrics that would be needed to evaluate predictive accuracy comprehensively.

Forecasting Accuracy Metrics

Forecasting accuracy is evaluated by comparing predicted paddy prices with the corresponding actual prices. Three commonly used regression metrics are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The coefficient of determination (\(R^2\)) can also be used to assess how well the model explains the variation in the evaluation targets.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between predicted and actual prices. A lower MAE indicates smaller average prediction errors.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Root Mean Squared Error (RMSE)

RMSE measures the square root of the average squared prediction errors. It penalizes larger errors more heavily than MAE.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Mean Absolute Percentage Error (MAPE)

MAPE measures the average absolute prediction error as a percentage of the actual values.

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

MAPE is suitable when actual target values are nonzero. Its interpretation also depends on whether the predictions and actual values are evaluated on the original price scale or a transformed scale.

XGBoost model evaluation metrics

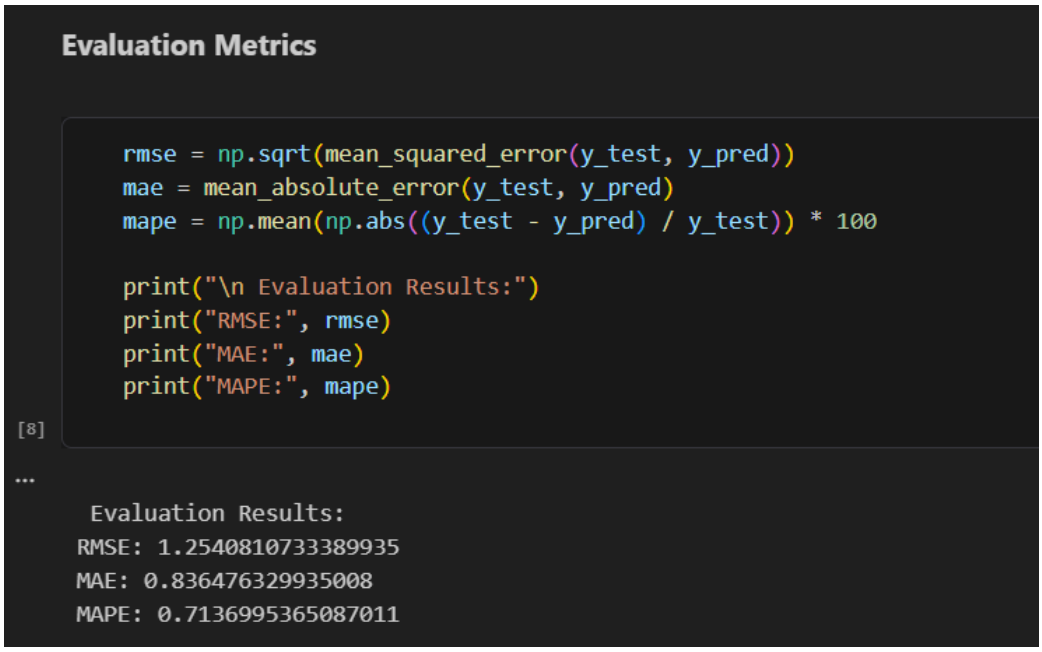


Figure 3 : XGBoost Model Evalution Metrics

API Response-Time Metrics

API performance is evaluated by measuring the time taken to process requests and return responses. Ten repeated requests were used for each of the three principal endpoints.

The following measures were recorded:

- Minimum response time: The fastest observed request.
- Maximum response time: The slowest observed request.
- Average response time: The arithmetic mean of the recorded request times.
- Median response time: The middle value after sorting the recorded times.
- Success rate: The percentage of requests that completed successfully.

3.5 Data analysis

The data analysis process evaluates the outputs generated by the proposed AI-driven paddy price forecasting and explainability component. It focuses on interpreting the results produced by the XGBoost prediction model, recursive multi-week forecasting mechanism, SHAP-based Explainable AI (XAI) layer, LLM explanation service, and FastAPI backend.

The analysis combines quantitative analysis of forecasting and API performance results with functional analysis of prediction explanations and system behaviour. The objective is to determine whether the developed component fulfils its intended functionality and how its measured performance relates to the research objectives.

The analysis is organized into five main areas:

1. Forecasting output and prediction accuracy.
2. SHAP-based feature contribution analysis.
3. XAI and LLM explanation analysis.
4. API performance and reliability analysis.
5. Functional testing and defect analysis.

Analysis of Forecasting Results

The forecasting results are analyzed to assess the behaviour of the XGBoost model when predicting paddy prices for selected districts and dates. The analysis considers both individual predictions and recursive multi-week forecasts.

For individual predictions, the predicted price is examined alongside the corresponding input district and date. For multi-week forecasting, the predicted prices are analyzed across the requested forecast horizon to identify the direction and pattern of the predicted price movement.

Where actual paddy prices are available for the same dates, the predicted values can be compared with the actual observations using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Analysis of SHAP Feature Contributions

SHAP outputs are analyzed to understand how individual features contribute to the predictions generated by the XGBoost model.

The analysis examines the selected features, their input values, and their SHAP contribution values. Features with larger absolute SHAP contributions have a greater influence on the particular prediction relative to the model's baseline, subject to the model and explainer configuration.

Analysis of XAI and LLM Explanations

The analysis of the explanation layer focuses on whether the structured evidence generated by the forecasting and XAI components is successfully transformed into understandable output.

The XAI modules generate the trend, confidence, risk level, market outlook, recommendation, and prediction summary. The LLM explanation service then uses the structured evidence to produce explanations in English and simplified Sinhala.

Analysis of API Performance

API performance is analyzed using the response times collected from repeated requests to the three principal endpoints: /predict, /forecast, and /explanation/llm.

The analysis uses the minimum, maximum, average, and median response times to characterize both typical performance and variation between requests.

3.6 Ethics privacy security and responsible AI

The development of the proposed Sri Lankan paddy price forecasting system considered ethical use, privacy, security, and responsible artificial intelligence (AI). Since the system generates price forecasts and explanatory information that may support decisions by farmers and other paddy supply chain stakeholders, it is important that its predictions are presented transparently, its data and services are protected, and its outputs are not treated as guaranteed market outcomes.

The system combines an XGBoost forecasting model, SHAP-based feature attribution, additional explainable AI (XAI) modules, and a large language model (LLM) to generate natural-language explanations. These components have different responsibilities. The XGBoost model produces the numerical price prediction, SHAP explains the contribution of features to the model output, and the XAI and LLM components help communicate the prediction and its supporting evidence. The LLM is intended to explain structured evidence rather than independently generate or modify the numerical forecast.

Ethical Considerations

Ethical considerations focus on ensuring that the system provides useful information without misleading users or encouraging them to rely on predictions without considering other relevant factors.

The system is designed to support decision-making rather than replace the judgement of farmers, agricultural professionals, or market participants. Forecasts represent estimates based on the model and available historical data. Actual paddy prices may differ because of market conditions, weather events, policy changes, transportation costs, and other factors that may not be fully represented in the dataset.

Security Considerations

Security is important because the forecasting system exposes functionality through REST API endpoints and loads a serialized machine-learning model and a historical dataset. Unauthorized access, invalid requests, configuration errors, or compromised dependencies could affect service availability or the integrity of the system.

The implementation includes centralized exception handling and request validation through FastAPI schemas. District validation is also performed before historical feature generation, helping the system return a meaningful error for unsupported districts rather than proceeding with an invalid forecasting context.

3.7 Reproducibility and research artefacts

Reproducibility is an important consideration in the development and evaluation of the proposed Sri Lankan paddy price forecasting system. It enables the research process, model behaviour, API functionality, and evaluation outcomes to be examined using the artefacts produced during development and testing. For this research component, reproducibility is supported by maintaining the forecasting model, historical dataset, source code, configuration files, API schemas, testing procedures, and recorded test results.

The system is implemented using Python and FastAPI, with an XGBoost model for price prediction, SHAP for feature attribution, additional XAI modules for interpreting prediction results, and an LLM-based service for generating natural-language explanations in English and simplified Sinhala. Keeping these components and their

responsibilities identifiable helps distinguish the numerical forecasting process from the subsequent interpretation and explanation stages.

Code and Model Reproducibility

The application separates data loading, feature generation, model inference, forecasting, explanation, and API handling into dedicated modules. This modular structure makes it easier to inspect individual processing stages and repeat relevant tests without treating the complete application as a single indivisible component.

The serialized model and historical dataset are particularly important to reproducibility. Repeating a prediction requires access to the corresponding model artefact, compatible software dependencies, and the same relevant input values. Where feature generation depends on historical records, the dataset and date used for the request must also be controlled.

Reproducibility Limitations

Several factors may affect the extent to which another researcher can reproduce the reported results.

First, the forecasting system depends on the availability and integrity of the historical dataset and trained model. Changes to either artefact may produce different forecasts. Second, recursive forecasting can propagate earlier prediction errors across the requested horizon. Third, LLM-generated explanations may vary depending on the provider, model version, prompt, and service configuration. Finally, performance measurements may differ across computers, network conditions, and software environments.

To improve reproducibility, the final research repository should include a dependency specification, documented execution and testing commands, representative API request and response examples, and a record of the model and dataset versions used for the reported experiments. Sensitive credentials and API keys should not be included in the public repository. Where model-training code and the complete training configuration are available, these should also be retained to support reproduction of the model artefact rather than only its inference behaviour.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements of the proposed Sri Lankan Paddy Price Forecasting System were identified based on the need to provide district-level paddy price predictions, multi-week forecasting, interpretable prediction results, and accessible explanations for users. The system was designed as a research component of the wider Digital Goviya ecosystem, with a focus on combining machine learning-based forecasting and explainable artificial intelligence (XAI) to support informed decision-making within the Sri Lankan paddy supply chain.

The requirements cover the functional capabilities of the forecasting service, its performance and reliability expectations, and the technical constraints associated with integrating machine learning models and language-based explanations through a REST API. They also consider the need to communicate forecast results clearly without presenting model predictions as guaranteed future market prices.

Requirements are organized into functional requirements, which describe what the system should do, and non-functional requirements, which describe the expected quality and operational characteristics of the system.

Functional Requirements

Functional requirements define the core services provided by the forecasting system, from accepting prediction requests to returning numerical forecasts and explanatory information.

Non-Functional Requirements

Non-functional requirements describe the expected performance, reliability, maintainability, and usability of the forecasting service. These requirements are particularly important because the system integrates computational forecasting, feature attribution, and an external language-model service, each of which can have different processing costs.

4.2 Component architecture within the complete solution

Overview

The Sri Lankan Paddy Price Forecasting System is designed as a modular research component within the wider Digital Goviya ecosystem. Its primary responsibility is to generate district-based paddy price predictions, produce multi-week forecasts, interpret model outputs using explainable AI (XAI), and provide natural-language explanations in English and simplified Sinhala.

The component follows a layered architecture in which the client application communicates with a FastAPI-based REST API, which coordinates the underlying data-processing, forecasting, explainability, and language-generation services. The architecture separates numerical prediction from interpretation and explanation so that the LLM communicates evidence generated by the forecasting and XAI components rather than independently determining the predicted price.

At the complete-solution level, the forecasting component acts as a specialized service that can be consumed by the wider Digital Goviya mobile application. This separation allows the forecasting logic to be developed, tested, and maintained independently of the client interface.

High-Level Component Architecture

The component architecture illustrates the interaction between the Digital Goviya client application and the forecasting backend, including the main internal services responsible for generating predictions and explanations.

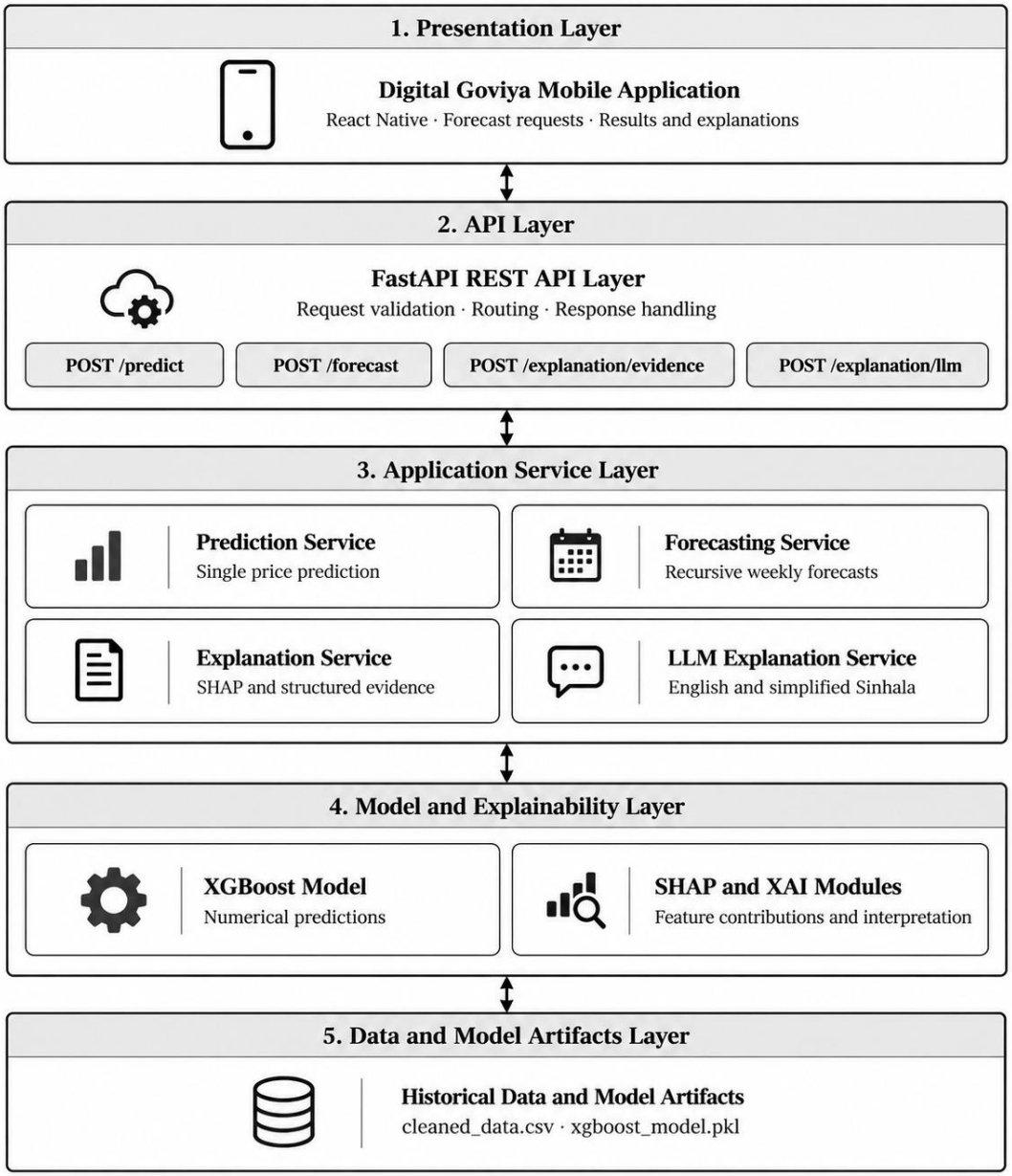


Figure 4 : Component architecture of the paddy price forecasting system within the Digital Goviya solution.

Integration Within the Complete Digital Goviya Solution

The forecasting component is intended to operate as a specialized backend service within the wider Digital Goviya ecosystem. Its REST API provides a boundary between the mobile application and the machine-learning implementation.

This arrangement offers three main architectural benefits:

1. Separation of concerns: The mobile interface, forecasting logic, and explanation generation are maintained as distinct responsibilities.
2. Independent testing and maintenance: The forecasting backend can be tested through API requests without requiring every test to be executed through the mobile interface.
3. Integration flexibility: Other compatible components of the wider solution can consume the forecasting service through its defined API interfaces.

The architecture therefore connects numerical prediction with interpretable evidence and user-facing explanations while preserving a clear distinction between the forecasting model and the language-generation component.

4.3 Design alternatives and decisions

The design of the Sri Lankan Paddy Price Forecasting System involved selecting suitable technologies and architectural approaches for price prediction, multi-week forecasting, explainability, and natural-language explanation generation. The decisions were guided by the research objectives, the available historical paddy price data, the need to provide interpretable predictions, and the requirement to integrate the forecasting component with the wider Digital Goviya application.

The selected solution combines an XGBoost forecasting model, SHAP-based feature attribution, additional XAI modules, a FastAPI REST API, and an LLM-based explanation service. Alternative approaches were considered at the conceptual design level, and the selected approaches were aligned with the implemented system.

The following subsections describe the principal design alternatives and explain the rationale for the selected approaches. The comparisons focus on the requirements of this research component rather than claiming that one technology is universally superior to all alternatives.

Forecasting Model Selection

A central design decision was selecting a machine-learning approach capable of learning relationships between historical paddy prices and the available input features.

Possible alternatives included statistical time-series models, deep-learning models, and tree-based machine-learning models.

Relationship Between Design Decisions

The selected approaches form a connected workflow rather than a collection of independent technologie

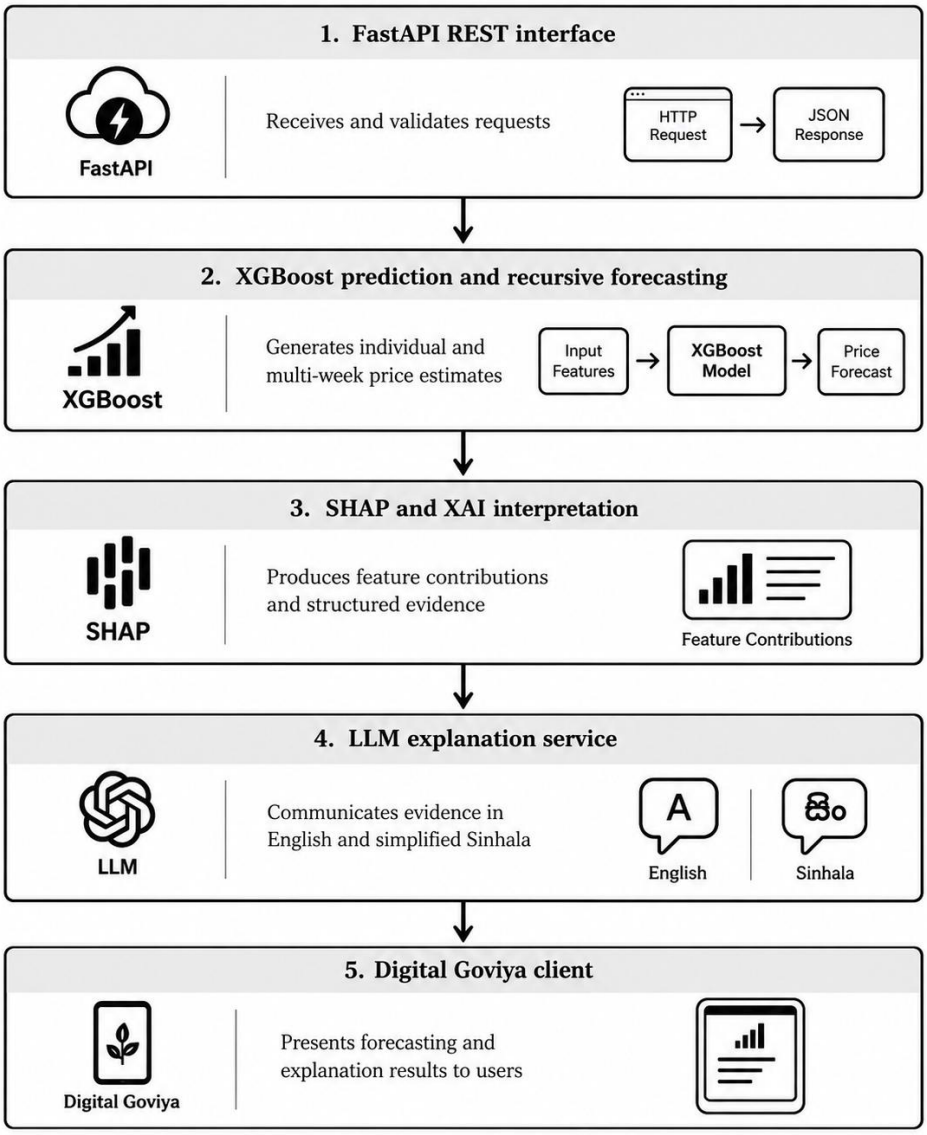


Figure 5 : Relationship between the selected design approaches in the forecasting solution.

4.4 Detailed design

The detailed design of the Sri Lankan Paddy Price Forecasting System describes how the selected architecture is translated into implementable software components, their responsibilities, their interactions, and the processing logic used to generate price predictions and explanations.

The system is implemented as a modular Python application using FastAPI for REST API communication, XGBoost for numerical price prediction, SHAP for feature attribution, additional explainable AI (XAI) modules for interpreting results, and a large language model (LLM) service for generating explanations in English and simplified Sinhala.

The detailed design focuses on five principal areas:

1. API and request-processing design.
2. Data loading, feature preparation, and model inference.
3. Recursive multi-week forecasting.
4. Prediction explanation and LLM-based interpretation.
5. Error handling, response validation, and integration with the Digital Goviya mobile application.

These areas are described below using the existing project structure and known implementation behaviour.

Data Loading and Feature Preparation

The data-processing design provides the information required by the forecasting model. The prepared historical dataset is stored in app/data/cleaned_data.csv, while the trained model is stored in app/models/xgboost_model.pkl.

The file paths are defined centrally in app/core/config.py, allowing the application to reference the model and dataset through a consistent configuration.

Detailed Prediction Design

The prediction component generates a numerical paddy price estimate for a supported district and specified date. It combines the prepared historical information, feature-generation logic, and trained XGBoost model.

The principal processing steps are:

1. Receive a prediction request through POST /predict.
2. Validate the request and check whether the selected district is supported.
3. Retrieve the historical data required for the prediction.
4. Prepare the model input features.
5. Load or access the trained XGBoost model through the model-loading mechanism.
6. Generate the numerical price prediction.
7. Invoke the explanation workflow where required by the endpoint.
8. Return the prediction and the corresponding response information.

Recursive Multi-Week Forecasting Design

The forecasting service extends the prediction functionality to multiple future weeks. Rather than requiring a separate trained model for each future week, the implemented approach reuses the trained prediction model recursively.

The process begins with a district, a starting date, and a requested forecast horizon. The service prepares the required features, predicts the next week's price, and appends a forecast record to the data used for subsequent iterations.

The appended record contains the forecast date, district, predicted average price, derived minimum and maximum price fields, and the previous record's `production_total` value. The minimum and maximum fields are generated as the predicted price minus and plus 2, respectively; these values are part of the current implementation and should not be interpreted as statistically estimated prediction intervals.

Explainability and XAI Design

The explainability subsystem is designed to provide information that helps users interpret the numerical prediction. It combines SHAP-based feature attribution with application-specific XAI modules.

The principal modules include:

- trend.py — categorizes price movement into predefined trend categories.
- confidence.py — assigns a high, medium, or low confidence category using available SHAP evidence.
- feature_importance.py — processes feature-importance information.
- recommendation.py — generates market outlook, risk, and recommendation information.
- summary.py — produces summary information from the available explanation evidence.

The explanation service coordinates relevant processing and creates structured evidence for the LLM explanation stage.

LLM-Based Explanation Design

The LLM explanation component converts structured evidence into a natural-language response intended to be easier for users to understand. The system supports English and simplified Sinhala.

The explanation workflow uses the following general sequence:

1. Obtain the structured evidence produced by the explanation service.
2. Prepare the evidence for the LLM explanation service.
3. Send the relevant evidence through the LLM client.
4. Generate explanations in the supported languages.
5. Structure the output according to the expected response format.
6. Return the explanation to the calling endpoint.

The structured evidence can include the district, date, predicted price, previous price, trend, confidence category, risk, outlook, recommendation, and relevant feature contributions.

The key design decision is that the LLM explains the evidence rather than replacing the forecasting model. The numerical prediction remains the responsibility of XGBoost.

API Response and Client Integration Design

The REST API acts as the communication boundary between the forecasting backend and the Digital Goviya client. The backend is responsible for model inference, forecasting, and explanation generation, while the client consumes the returned information and presents it through the mobile interface.

The API schemas define the structures expected by the relevant operations. Keeping these structures consistent helps the client interpret predictions, weekly forecast records, and explanation responses.

4.5 Standards and professional practice

The development of the Sri Lankan Paddy Price Forecasting System follows relevant software engineering practices for modular development, API design, data validation, testing, maintainability, and responsible use of artificial intelligence. These practices support the development of a system that can generate paddy price forecasts, provide interpretable evidence, and communicate results through the wider Digital Goviya application.

The system integrates several technologies, including Python, FastAPI, XGBoost, SHAP, and an LLM-based explanation service. Since these components perform different functions, consistent interfaces, appropriate validation, clear separation of responsibilities, and documented testing are important throughout the development process.

This section identifies the relevant standards and professional practices associated with the solution. It distinguishes practices reflected in the implementation from broader standards that provide guidance for further evaluation and improvement.

Software Engineering Standards and Practices

The system uses a modular architecture that separates API routing, data loading, model inference, forecasting, explanation generation, and error handling. This design

supports separation of concerns and makes it easier to inspect and test individual components.

Relevant International Standards and Frameworks

Several international standards and frameworks provide useful guidance for the design and evaluation of the proposed system. These are used as reference points for professional practice; their relevance does not imply formal certification or full compliance.

API Design and Integration Practices

The forecasting backend uses FastAPI to expose prediction and explanation functionality through REST API endpoints. The API design emphasizes clear endpoint responsibilities, request validation, structured responses, and controlled error handling.

The following practices are relevant to API development:

- Clear endpoint responsibilities: Each endpoint provides a defined capability, such as single prediction, recursive forecasting, or explanation generation.
- Schema-based validation: Defined request and response schemas support consistent data exchange between the client and backend.
- Appropriate error responses: Invalid inputs and processing failures should be communicated without unnecessarily exposing internal implementation details.
- Independent testing: API endpoints can be tested through automated tests or API clients without requiring interaction through the mobile application.
- Separation of client and server logic: Numerical prediction and explanation generation remain within the backend, while the client presents the returned results.

These practices support integration with the Digital Goviya mobile application and help maintain consistency as the forecasting component evolves.

Testing and Quality Assurance Practices

Testing is an important part of the professional development process because successful execution alone does not establish that a forecasting system behaves correctly under all conditions.

The research component was evaluated using unit tests, integration tests, system tests, defect retesting, and non-functional performance and reliability measurements.

Testing practice	Purpose in the system
Unit testing	Examines individual XAI modules, including trend, confidence, feature importance, and recommendation logic.
Integration testing	Examines interactions between components, including model loading, forecasting, and explanation services.
System testing	Verifies API behaviour using valid and invalid requests.
Regression testing	Checks whether changes, such as the unsupported-district validation fix, affect existing functionality.
Performance testing	Measures response times for prediction, forecasting, and LLM explanation endpoints.
Reliability testing	Records successful and failed requests during repeated-request tests.
Defect documentation	Records observed issues, their corrections, and supporting before-and-after evidence.

Table 8 : Testing Practices and Their Purposes

Security, Privacy, and Responsible AI Practices

Professional practice also requires attention to security, privacy, and responsible use of machine-learning outputs. These concerns are particularly relevant because the system uses historical agricultural data and an external LLM service.

The design incorporates input validation, controlled error handling, separation of prediction and explanation responsibilities, and awareness of the limitations of model-generated outputs. Further operational safeguards should be verified according to the deployment environment.

The main considerations are summarized below.

- Data protection: Use the historical information necessary for forecasting and avoid transmitting unnecessary personal information to external services.
- Credential management: Keep API keys and other secrets out of source code and protect them through appropriate deployment configuration.
- Explanation integrity: Ensure that natural-language explanations remain consistent with the structured prediction evidence.
- Transparency: Communicate that forecasts and recommendations are estimates and decision-support information, not guaranteed market outcomes.
- Dependency maintenance: Review software dependencies and address relevant security vulnerabilities.
- Human judgement: Encourage users to consider forecasts alongside current market information and their own context.

These practices distinguish responsible design intentions from verified implementation controls. For example, the existence of centralized exception handling does not establish that the system has undergone a complete security audit.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The Sri Lankan Paddy Price Forecasting System was developed as a Python-based backend component within the wider Digital Goviya solution. The development environment and technology stack were selected to support machine-learning inference, historical data processing, recursive multi-week forecasting, explainable artificial intelligence (XAI), and large language model (LLM)-based explanation generation.

The implementation follows a modular architecture in which the REST API, data-loading logic, forecasting services, explanation modules, and external language-model integration are maintained as separate components. This structure supports independent development and testing while allowing the forecasting functionality to be accessed by the Digital Goviya mobile application.

The technology stack consists of Python for the core implementation, FastAPI for API development, XGBoost for numerical price prediction, SHAP for feature attribution, and an LLM service for generating explanations in English and simplified Sinhala. The historical dataset and trained model are maintained as separate artefacts and loaded by the relevant application services.

Development Environment

The development environment provides the tools and runtime required to implement, execute, debug, and test the forecasting backend. The environment supports the development of individual services and the verification of their interactions through API requests and automated tests.

The development environment supported iterative implementation and verification. Individual services could be examined through automated tests, while Postman could be used to submit requests directly to the running API. This enabled the backend to be tested independently of the mobile application's user interface.

The exact Python version, operating system version used during the experiments, and dependency versions should be recorded from the actual development and test

environment. These details are important for reproducing the implementation and should not be estimated in the final thesis.

Technology Stack

The technology stack combines machine-learning, explainability, API development, and language-generation technologies. Each technology serves a specific purpose within the forecasting workflow.

Technology	**Role in the system**	**Implementation**
Python	Core backend development	Implements data processing, prediction, forecasting, and explanation services.
FastAPI	REST API framework	Exposes the system's functionality through defined HTTP endpoints.
XGBoost	Machine-learning model	Generates numerical paddy price predictions using the prepared model inputs.
SHAP	Model explainability	Estimates the contribution of input features to individual predictions.
Pandas	Historical data processing	Supports the manipulation of structured historical data, where used by the data-processing implementation.
Pydantic	Data validation and schemas	Supports request validation and structured API data contracts through FastAPI.
LLM service	Natural-language generation	Converts structured prediction evidence into English and simplified Sinhala explanations.
CSV	Historical data storage	Stores the prepared historical dataset in app/data/cleaned_data.csv.

Pickle-based model artefact	Model persistence	Stores the trained model in app/models/xgboost_model.pkl.
React Native	Client application integration	Provides the mobile application through which forecasting functionality can be accessed.

Table 9 : Technology Stack of the Proposed System

5.2 Component implementation

The component implementation describes how the design of the Sri Lankan Paddy Price Forecasting System was translated into working software modules. The implementation focuses on the development of the prediction API, historical data processing, XGBoost model integration, recursive multi-week forecasting, SHAP-based explainability, additional XAI modules, and LLM-based explanation generation.

The backend was implemented using Python and FastAPI, with each major responsibility organized into a separate module. This modular approach supports independent testing and simplifies the integration of the forecasting functionality with the Digital Goviya mobile application.

The following subsections describe the main implemented components, their responsibilities, and their interactions. The descriptions are aligned with the established project structure and known implementation behaviour.

The components are integrated through service calls and the REST API rather than being implemented as completely independent applications.

The principal integration relationships are summarized below.

Source component	**Integrated component**	**Purpose**
API routes	Prediction service	Processes individual prediction requests.
API routes	Forecasting service	Processes multi-week forecasting requests.

Data loader	Feature service	Provides historical records required for feature preparation.
Model loader	Prediction and forecasting services	Provides access to the trained XGBoost model.
Explanation service	SHAP and XAI modules	Generates structured interpretation evidence.
LLM explanation service	LLM client	Obtains natural-language explanations from the configured language-model service.
API layer	Digital Goviya mobile application	Exposes forecasting and explanation functionality to the client.

Table 10 : Principal component integration relationships.

5.3 Integration with the other components

The Sri Lankan Paddy Price Forecasting System is developed as a specialized backend component within the wider Digital Goviya ecosystem. Its integration design enables the forecasting functionality to communicate with the mobile application and coordinate the internal services responsible for data processing, prediction, explainability, and natural-language explanation generation.

The system uses FastAPI to expose REST API endpoints, allowing the client application to submit prediction and forecasting requests and receive the corresponding results. Internally, the backend integrates the historical dataset, trained XGBoost model, feature preparation services, SHAP-based explainability, additional XAI modules, and the LLM explanation service.

Integration is organized at two levels: internal integration, which coordinates the forecasting backend's services, and external integration, which connects the forecasting backend to the Digital Goviya mobile application and the configured LLM provider.

Internal Component Integration

Internal integration connects the modules within the forecasting backend so that each component can perform its assigned responsibility while exchanging the required data with other components.

The data-loading service provides access to the historical paddy price dataset, and the feature service prepares the model inputs. The prediction and forecasting services use the trained XGBoost model to generate individual price predictions and multi-week forecasts. The explanation service coordinates SHAP-based feature attribution and the additional XAI modules to generate structured evidence. When a natural-language explanation is requested, the LLM explanation service uses that evidence to generate an English or simplified Sinhala response.

Integration with the Digital Goviya Mobile Application

The forecasting backend is integrated with the Digital Goviya mobile application through REST APIs. The mobile application provides the user interface, while the backend performs the numerical prediction, multi-week forecasting, and explanation-generation operations.

The client sends requests containing the parameters required by the relevant endpoint. The FastAPI backend validates the requests, invokes the appropriate services, and returns structured responses. The client can then present the results to the user through its interface.

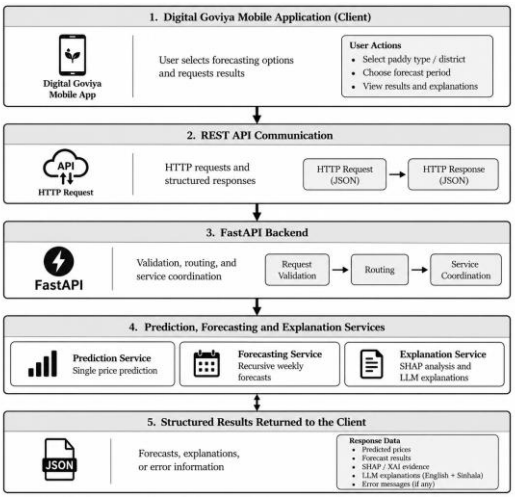


Figure 6 : Integration between the forecasting backend and the Digital Goviya mobile application.

Integration Testing and Verification

Integration testing was used to verify the interactions between the backend's main components and API endpoints. The tests examined whether model loading, forecasting, structured evidence generation, and LLM explanation generation operated through their expected interfaces.

Integration area	Verification activity	Recorded evidence or status
Model loading	Test the loading and availability of the trained model.	Model-loading integration test recorded as passed.
Forecasting	Execute multi-week forecast requests and verify the returned response structure.	Two forecasting integration tests recorded as passed.
Structured explanation	Request evidence through /explanation/evidence.	Integration test recorded as passed.
LLM explanation	Request a natural-language explanation through /explanation/llm.	Integration test recorded as passed.
Error handling	Submit invalid or unsupported inputs and examine the returned error behaviour.	Unsupported-district correction documented through before-and-after evidence.
Client integration	Verify that the backend exposes the interfaces needed by the client.	API interfaces are implemented; end-to-end verification should be supported by the corresponding client integration tests.

Table 11 : Integration Verification

Integration Challenges and Resolutions

During development, integration testing helped identify issues in the interaction between input validation, historical data processing, and API responses.

One identified issue occurred when an unsupported district was processed before its validity had been checked. The application could return an insufficient-history error instead of identifying the unsupported district. The validation sequence was corrected so that district support is checked before historical feature generation.

Another integration consideration is the dependency on the LLM service. The explanation workflow depends on successful communication with the configured provider, so the application includes fallback behaviour for relevant failures. This separates the reliability of the numerical forecasting component from the availability of natural-language generation.

These corrections and safeguards improve the clarity and robustness of component interactions, although they do not eliminate every possible integration or deployment failure.

5.4 Verification and software testing

Verification and software testing were conducted to assess whether the Sri Lankan Paddy Price Forecasting System behaved according to its functional requirements and implemented design. The testing process focused on the correctness of individual components, the integration of forecasting and explanation services, the behaviour of API endpoints under valid and invalid inputs, and the performance and reliability of the backend under the tested conditions.

The testing activities covered the XGBoost model-loading process, recursive multi-week forecasting, SHAP-based explanation evidence, LLM-generated explanations, input validation, error handling, and API response times. Automated tests and direct API requests were used to examine component behaviour and identify implementation defects.

The evaluation included unit testing, integration testing, system testing, defect verification, and non-functional testing. These activities provided evidence of the system's observed behaviour while distinguishing functional correctness and operational performance from predictive accuracy, which requires a separate evaluation against actual paddy prices.

Verification Approach

Verification was organized into several testing levels, with each level addressing a different aspect of the implementation.

Testing level	Objective	Main components or activities
Unit testing	Verify the behaviour of individual modules.	Trend, confidence, feature importance, recommendation, and summary logic.
Integration testing	Verify interactions between related components.	Model loading, forecasting, structured explanation evidence, and LLM explanation generation.
System testing	Verify the behaviour of the application through its API interfaces.	Valid requests, unsupported districts, insufficient historical data, and invalid forecast horizons.
Regression testing	Check whether a code change affects previously working functionality.	Rechecking the system after correcting the unsupported-district validation defect.
Performance testing	Measure request-processing time under specified test conditions.	Prediction, forecasting, and LLM explanation endpoints.
Reliability testing	Observe successful and failed requests during repeated execution.	Repeated prediction requests and sequential request tests.

Table 12 : Verification and Testing Levels

5.5 Deployment and operation

The deployment and operation of the Sri Lankan Paddy Price Forecasting System involve making the implemented forecasting backend available for API requests, configuring the required model and data resources, and ensuring that its prediction, forecasting, and explanation services can operate as intended.

The backend is implemented using Python and FastAPI. It depends on the prepared historical paddy price dataset, the trained XGBoost model, the associated XAI

modules, and the configured LLM service for natural-language explanation generation. The system exposes REST API endpoints that allow the Digital Goviya mobile application and API testing tools to access the forecasting functionality.

Operational activities include starting the application, checking endpoint availability, submitting prediction and forecasting requests, handling invalid inputs, and monitoring response behaviour. Deployment configuration must also account for dependencies and credentials required by the application.

The following subsections describe the deployment arrangement and operational workflow based on the implemented system. Specific hosting infrastructure and production deployment procedures should only be included where they are supported by the actual deployment configuration.

Deployment Architecture

The forecasting backend is deployed as a Python-based API service that separates the client application from the machine-learning processing logic. The FastAPI application exposes endpoints for prediction, recursive forecasting, structured explanation evidence, and LLM-based explanation generation.

The deployment arrangement consists of the following main elements:

- Client application: The Digital Goviya React Native mobile application, which sends requests to the backend.
- API service: The FastAPI application, which receives requests and coordinates the relevant services.
- Model and data resources: The serialized XGBoost model and prepared historical dataset required for inference.
- Explanation services: The SHAP and XAI modules that generate supporting evidence for the predictions.
- LLM service: The configured external service used to generate natural-language explanations.

Operational Workflow

The operational workflow begins when a user requests a prediction or forecast from the mobile application. The backend validates the request and executes the relevant processing pipeline. The response is then returned to the client for presentation.

The main operational stages are:

1. Receive a prediction, forecasting, or explanation request.
2. Validate the request parameters.
3. Access the historical dataset and trained model when required.
4. Execute the relevant prediction or forecasting service.
5. Generate structured explanation evidence when requested.
6. Invoke the LLM service when a natural-language explanation is required.
7. Return the result or an appropriate error response.
8. Record and investigate failures where operational logging is configured.

The system also validates supported districts before historical feature generation. This helps prevent incorrect error classification and provides clearer feedback when users submit unsupported inputs.

Deployment Security and Maintenance

The deployment environment must protect the model artefact, historical dataset, application configuration, and any credentials required by the LLM service. API keys and other secrets should be managed through suitable environment or deployment configuration rather than committed to the source-code repository.

The deployed application should also use compatible dependency versions. This is particularly relevant to the serialized XGBoost model because a model created under one software version may produce warnings or compatibility problems when loaded under another version.

Maintenance activities should include reviewing dependencies, verifying model and dataset availability, checking API behaviour after updates, and repeating relevant tests following code or configuration changes. Where the model or dataset is updated, the

resulting forecasts should be reevaluated before the updated version is used for decision support.

6 RESULTS AND VALIDATION

6.1 Research results

This section presents the results obtained from implementing and evaluating the Sri Lankan Paddy Price Forecasting System. The evaluation focuses on the system's ability to generate district-level paddy price predictions, produce recursive multi-week forecasts, provide explainable prediction evidence, generate natural-language explanations, and process API requests reliably under the tested conditions.

The results are organized around the main objectives of the research component. These include the implementation of the XGBoost-based forecasting service, integration of SHAP and additional explainable AI (XAI) modules, generation of English and simplified Sinhala explanations using a large language model (LLM), and evaluation of the backend's functional behaviour and response performance.

The findings presented in this section distinguish between directly recorded test results, illustrative forecast outputs, and aspects that require additional empirical validation. In particular, successful API execution demonstrates that the implemented functionality operates under the tested conditions; it does not, by itself, establish that the forecasting model achieves a particular level of predictive accuracy or improves farmers' decisions.

Results of Price Prediction and Multi-Week Forecasting

The implemented system successfully generated numerical paddy price predictions through the XGBoost model and supported multi-week forecasting through the recursive forecasting service. The forecasting service repeatedly prepared model inputs, generated a prediction for the next week, and incorporated that prediction into the data used for the subsequent forecast step.

A recorded forecast request for Anuradhapura, starting on 31 August 2026, generated six weekly predictions.

The recorded output shows that the predicted values remained relatively stable over the six-week forecast horizon. The first predicted value was 112.07, while the final

predicted value was 111.91. This illustrates the recursive forecasting functionality and the pattern produced by the model for this particular request.

However, the stability of these predictions should not be interpreted as evidence of forecasting accuracy. A separate comparison against actual paddy prices for the corresponding dates is required to determine how closely the predictions represent observed market conditions.

Results of Explainability and Interpretation

The system integrates SHAP-based feature attribution with additional XAI modules to provide information supporting the interpretation of individual predictions. The implemented modules cover trend classification, confidence categorization, feature importance, recommendations, and summary generation.

The results of these components are intended to extend the numerical prediction with structured interpretation evidence.

Component	**Result produced**	**Role in the research**
SHAP-based explanation	Feature contribution values	Helps interpret how input features contribute to an individual model prediction.
Trend analysis	Predefined price trend category	Communicates the interpreted direction and strength of price movement.
Confidence assessment	High, medium, or low category	Provides an application-specific assessment based on available SHAP evidence.
Feature-importance module	Feature-importance information	Supports interpretation of relevant features.
Recommendation module	Market outlook, risk, and recommendation	Provides decision-support information based on the implemented logic.

Summary module	Summary of available evidence	Supports concise interpretation of the results.

Table 13 : Explainability and Interpretation Capabilities

These components demonstrate that the forecasting system provides an explanation workflow in addition to numerical inference. SHAP values describe model behaviour rather than causal relationships, and the confidence category is not a calibrated probability of prediction correctness.

Although these capabilities are implemented, claims about explanation accuracy, consistency, or usefulness to farmers require separate evaluation evidence.

Results of LLM-Based Explanation Generation

The system integrates an LLM explanation service that transforms structured prediction and XAI evidence into natural-language explanations in English and simplified Sinhala. The explanation workflow receives information such as the district, prediction date, predicted price, previous price, trend, confidence category, risk, market outlook, recommendation, and relevant feature contributions.

The implementation provides separate API functionality for structured explanation evidence and LLM-based explanation generation. This separation allows the numerical prediction and its supporting evidence to be distinguished from the natural-language response.

Results of Functional and Integration Testing

The recorded testing activities evaluated individual XAI modules, interactions between backend services, and the behaviour of the system through its API endpoints.

Testing area	**Recorded result**
Trend module unit tests	6 tests passed in a recorded run.
Combined XAI unit tests	A recorded run covering recommendation, summary, feature importance, confidence, and trend modules passed 47 tests.

Model-loading integration test	1 test passed.
Forecasting integration tests	2 tests passed.
Structured explanation integration test	1 test passed.
LLM explanation integration test	1 test passed.
Unsupported-district validation	Corrected behaviour was verified through retesting.
Invalid forecast horizon	Validation for a zero-week forecast request was tested.

Table 14 : Summary of Recorded Software Testing Results

The recorded results provide evidence that the tested components and endpoints performed their expected functions in the corresponding test runs. The test counts represent individual runs and overlapping test scopes; they should not be added together to calculate a total number of unique tests.

The unsupported-district defect also demonstrated the value of system testing in identifying incorrect error classification. Following the correction, the application returned an error identifying the unsupported district rather than incorrectly reporting insufficient historical data.

A final regression-test result after this correction was not confirmed in the available record. Therefore, a successful post-fix regression run should be reported only after its output has been verified.

6.2 Product and technical results

This section presents the product-level and technical outcomes achieved through the development and integration of the proposed Sri Lankan paddy price forecasting component. The evaluation focuses on the operational forecasting service, explainable

artificial intelligence (XAI) functionality, natural-language explanation generation, API integration, and observed system performance.

The results demonstrate the implementation of a functional forecasting component capable of generating district-specific paddy price estimates and presenting supporting explanations. However, successful technical implementation and API testing should be distinguished from the statistical accuracy of the forecasting model and the effectiveness of the system in real-world agricultural decision-making.

Functional Product Outcomes

The developed forecasting component provides a set of interconnected capabilities intended to support paddy price analysis and interpretation. The component is implemented as a FastAPI-based backend service, designed to communicate with the Digital Goviya mobile application through RESTful APIs.

The primary product outcome is the integration of machine learning predictions with explainability and user-oriented interpretation within a single service architecture.

Forecasting Service Results

The forecasting service uses the trained XGBoost model to produce price predictions based on the features generated from available historical records. For multi-week forecasting, a recursive approach is used in which each generated prediction contributes to the data used for subsequent forecasting steps.

This design enables the service to generate a sequence of future weekly estimates rather than limiting the output to a single prediction.

A six-week forecast generated for Anuradhapura, beginning on 31 August 2026, demonstrated the operational behavior of the forecasting service. The predicted values remained within a relatively narrow range, indicating a stable forecast pattern for that particular input scenario.

The observed result confirms that the implemented service can generate a sequence of date-aligned weekly forecasts. Nevertheless, this example does not establish predictive accuracy because the corresponding actual future prices were not evaluated in the recorded test.

Explainable AI Results

A significant technical outcome of the component is the integration of explainability mechanisms with the forecasting model. Instead of returning only a numerical price estimate, the system generates additional evidence to support the interpretation of each prediction.

SHAP-based feature contributions provide information about how input features influence the model output. These contributions are processed by dedicated XAI modules to generate trend classifications, feature-importance information, confidence categories, risk assessments, and market-oriented summaries.

The explainability architecture separates the underlying prediction from its interpretation. The XGBoost model remains responsible for generating the numerical estimate, while the explanation modules process prediction-related evidence.

This separation is important because it prevents the interpretation layer from being treated as an independent forecasting mechanism.

Natural-Language Explanation Results

The integration of the large language model represents another product-level outcome. The explanation service receives structured evidence containing the predicted price, previous price, trend, confidence category, risk category, market outlook, recommendations, and relevant feature contributions.

Based on this information, the service generates human-readable explanations in English and simplified Sinhala.

This approach is intended to reduce the difficulty of interpreting technical forecasting information, particularly when the output is presented through a mobile application.

The explanation workflow was verified through integration testing of the structured evidence and LLM explanation endpoints. Both relevant integration tests passed in their recorded runs.

An additional technical outcome is the implementation of fallback behavior for the LLM explanation service, allowing the application to handle explanation-generation failures without treating them as failures of the underlying forecasting model.

Technical Performance Results

Performance testing was conducted for the prediction, forecasting, and LLM explanation endpoints using ten requests per endpoint.

API endpoint	Requests	Average (ms)	Median (ms)	Maximum (ms)	Successful
/predict	10	53.3	35.5	200	10/10
/forecast	10	105.7	93.0	179	10/10
/explanation/llm	10	371.0	315.5	910	10/10

Table 15 : API Response-Time Performance Results

6.3 User or stakeholder validation

User or stakeholder validation is important for assessing whether the developed Sri Lankan paddy price forecasting component addresses the practical information needs of its intended users. While technical testing evaluates whether the system operates according to its requirements, stakeholder validation examines whether the system's outputs are understandable, relevant, and potentially useful in agricultural decision-making.

The proposed forecasting component is intended to support the Digital Goviya ecosystem by providing district-specific paddy price estimates, future price trends, explanations of model predictions, and market-oriented recommendations. These outputs are designed to help users interpret forecasting information rather than rely exclusively on numerical predictions.

Important: The implementation and software testing results described in the previous sections do not, by themselves, demonstrate that farmers or other stakeholders have evaluated the system. The available project information does not confirm completed interviews, questionnaires, usability sessions, or stakeholder feedback. Therefore, this section distinguishes the intended validation criteria from the evidence currently available.

Intended Stakeholders

The component is designed to support stakeholders who need to interpret paddy price information within the Sri Lankan agricultural supply chain.

Stakeholder group	**Potential use of the component**	**Validation priority**
Paddy farmers	Review estimated future prices and market trends	Understandability, relevance, and usefulness for planning
Paddy collectors and traders	Interpret expected price movements	Clarity of forecasts and market outlook
Warehouse operators	Consider price information alongside storage and selling decisions	Relevance of forecasting periods and trends
Agricultural officers and advisors	Interpret price estimates and supporting explanations	Interpretability and responsible communication
Digital Goviya application users	Access forecasts through the mobile interface	Ease of use, response clarity, and accessibility

Table 16 : Intended Stakeholders and Their Validation Priorities

These groups represent intended stakeholder categories rather than confirmed research participants. Their inclusion identifies potential evaluation needs but does not establish that feedback was collected from them.

Stakeholder-Oriented Product Features

Several implemented features provide a foundation for future stakeholder evaluation.

First, the forecasting service generates district-specific predictions and supports multi-week forecasts. These functions can be evaluated to determine whether the selected geographic scope and forecasting horizon match stakeholder requirements.

Second, the XAI modules provide trend classifications, feature contributions, confidence categories, risk assessments, and recommendations. These outputs can be

assessed to determine whether users can interpret the prediction evidence without needing detailed knowledge of machine learning.

Third, the natural-language explanation service supports English and simplified Sinhala. This feature provides a basis for evaluating language accessibility among intended users. However, implementing two language outputs does not independently establish translation quality or user comprehension.

Finally, the component exposes its functionality through REST APIs for integration with the Digital Goviya mobile application. Evaluation of the complete user experience should therefore consider both the backend outputs and how those outputs are displayed and explained in the application.

Limitations of Stakeholder Validation

The principal limitation is the absence of documented empirical feedback from farmers, traders, agricultural advisors, or other intended users. Consequently, it is not currently possible to conclude that the component improves decision-making, increases user trust, or makes machine learning explanations easier to understand in practice.

The usefulness of a forecast also depends on its predictive accuracy and the relevance of its input data. Stakeholder perceptions of usefulness should therefore be interpreted alongside appropriate model-evaluation results rather than as a substitute for them.

In addition, confidence categories and market recommendations require careful evaluation to ensure that users do not interpret them as guarantees of future prices. Any stakeholder evaluation should communicate that predictions are estimates and that SHAP contributions explain model behavior rather than prove causal relationships.

6.4 Integration results

The integration of the Sri Lankan paddy price forecasting component focused on connecting the machine learning model, forecasting service, explainability modules, natural-language explanation service, and REST API layer into a unified backend component for the Digital Goviya ecosystem.

The primary objective of integration was to ensure that individual modules could exchange data correctly and operate together to deliver predictions and their associated explanations. The implemented system uses FastAPI to expose its functionality through API endpoints, allowing the React Native mobile application and other authorized components to request forecasting information without directly accessing the internal machine learning implementation.

Integration Architecture

The component follows a modular integration architecture in which each service performs a specific responsibility. Historical data and generated features are processed before being passed to the trained XGBoost model. The resulting prediction is then used by the explanation services to generate supporting evidence and, where requested, natural-language interpretations.

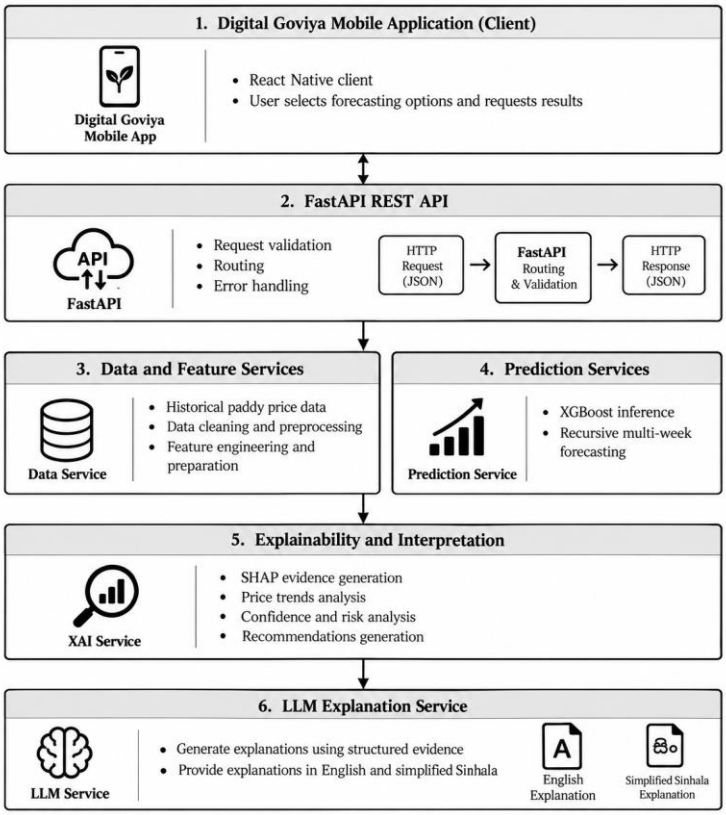


Figure 7 : Integration architecture of the paddy price forecasting component

Integration of Explainability Modules

The explanation service integrates SHAP-based feature contributions with dedicated XAI modules. These modules process prediction-related evidence to generate trend categories, confidence interpretations, feature-importance information, market outlooks, risk categories, and recommendations.

The structured evidence endpoint, /explanation/evidence, was tested independently to verify that the explanation evidence could be generated through the API. Its recorded integration test passed.

The combined architecture allows the prediction service and the explanation service to cooperate without requiring the model to generate natural-language output directly.

Integration of the LLM Explanation Service

The LLM explanation service was integrated to transform structured prediction evidence into human-readable explanations. The service receives relevant information from the explanation layer, including the predicted price, previous price, trend, confidence category, risk category, market outlook, recommendations, and important feature contributions.

The service then generates English and simplified Sinhala explanations based on the supplied evidence.

The integration was exposed through the /explanation/llm endpoint. The recorded integration test for this endpoint passed after the test was corrected to use the implemented route and service instance.

The implementation also includes fallback behavior for LLM explanation failures, allowing the explanation workflow to handle unsuccessful language-generation requests separately from numerical prediction processing.

Integration Error Handling and Reliability

Integration testing also helped identify and correct an error-handling issue. Initially, a request containing an unsupported district could be incorrectly interpreted as a lack of sufficient historical data because district validation did not occur before historical feature generation.

The validation sequence was corrected so that the selected district is checked before historical feature generation begins. Retesting confirmed that the service returned the appropriate unsupported-district message.

The backend also validates forecast duration, with a request specifying zero forecast weeks rejected by FastAPI validation because the permitted value must be at least one.

These cases demonstrate the importance of validating requests at the API boundary before invoking downstream processing services.

6.5 Summary of findings

This section summarizes the principal findings obtained from the development, implementation, testing, and integration of the Sri Lankan paddy price forecasting component. The findings are organized around the research objectives, focusing on forecasting functionality, explainability, natural-language interpretation, technical performance, and integration within the Digital Goviya ecosystem.

Key Research Findings

Finding 1: Successful implementation of district-specific paddy price forecasting

The developed component successfully integrates a trained XGBoost model with a FastAPI backend to generate district-specific paddy price estimates. The recursive forecasting service also generates consecutive weekly estimates, as demonstrated by the six-week Anuradhapura forecast beginning on 31 August 2026.

This finding establishes the operational feasibility of generating multi-week forecasts through the implemented service. However, the recorded forecast example alone does not establish the model's predictive accuracy against actual market prices.

Finding 2: Integration of explainability with numerical predictions

The component incorporates SHAP-based feature contributions and dedicated XAI modules to interpret forecasting outputs. These modules provide trend classifications, feature-importance information, application-defined confidence categories, risk assessments, market outlooks, and recommendations.

The recorded unit and integration tests provide evidence that these explanation components were implemented and tested. Nevertheless, the explanatory outputs should not be interpreted as proof of causal relationships or statistically calibrated prediction confidence.

Finding 3: Natural-language explanation generation in English and simplified Sinhala

The LLM explanation service transforms structured prediction evidence into human-readable explanations in English and simplified Sinhala. Its integration with the backend was verified through the recorded tests for structured evidence generation and LLM explanation generation.

This finding demonstrates the technical feasibility of combining machine learning outputs with natural-language interpretation. The extent to which these explanations improve comprehension among Sri Lankan farmers remains unverified because no confirmed user-comprehension evaluation results are available.

Finding 4: Functional integration of the backend services

The prediction, forecasting, explainability, and LLM explanation services were integrated through a modular FastAPI architecture. The API endpoints provide a consistent interface for requesting forecasting results and associated explanations.

Integration testing also identified an unsupported-district validation issue, which was corrected and subsequently retested. This demonstrates how systematic testing contributed to improving the correctness of error reporting.

Finding 5: Promising observed response times in the test environment

The recorded performance tests produced the following average response times:

- Prediction endpoint: 53.3 ms
- Forecasting endpoint: 105.7 ms
- LLM explanation endpoint: 371.0 ms

The prediction endpoint recorded the lowest average response time, while the LLM explanation endpoint recorded the highest. These measurements provide initial

evidence of the operational performance of the implemented services under the tested conditions, rather than a guarantee of production performance.

Finding 6: Successful repeated-request execution in recorded tests

The reliability tests recorded 20 successful responses from 20 consecutive prediction requests and 30 successful responses from 30 sequential repeated prediction requests. Both tests recorded an observed success rate of 100%.

These findings indicate stable execution within the tested scenarios. They do not establish production availability or concurrent multi-user capacity, because the 30-request test was sequential rather than concurrent

Summary of Results Against Research Objectives

Research objective area	**Main finding**	**Evidence status**
Develop district-specific price forecasting	Prediction service and recursive weekly forecasting implemented	Functionality tested
Integrate explainable AI techniques	SHAP-based evidence and XAI interpretation modules implemented	Unit and integration tests recorded
Generate human-readable explanations	English and simplified Sinhala explanation generation integrated	LLM integration test passed
Integrate forecasting functionality through APIs	FastAPI endpoints expose prediction, forecasting, and explanation services	Selected endpoints tested
Evaluate technical performance and reliability	Response times and repeated-request success measured	Quantitative test results available

Assess practical stakeholder usefulness	Intended user-facing features identified	Direct stakeholder validation not confirmed
Establish predictive accuracy	Forecast outputs generated	Held-out accuracy metrics not established in the available evidence

Table 17 : Summary of Research Objective Achievement

7 DISCUSSION

7.1 Interpretation of findings

The findings of this research demonstrate the technical feasibility of developing a Sri Lankan paddy price forecasting component that integrates machine learning, explainable artificial intelligence (XAI), and natural-language explanation generation within a unified backend service. The results indicate that the developed component can generate district-specific price estimates, produce recursive multi-week forecasts, interpret prediction-related evidence, and expose these capabilities through RESTful APIs. However, the findings must be interpreted in relation to the scope of the evaluation, distinguishing successful software implementation from demonstrated predictive accuracy and real-world usefulness.

Interpretation of Forecasting Results

The successful execution of the prediction and forecasting services indicates that the trained XGBoost model was integrated into a functional forecasting workflow. The recursive forecasting mechanism further enables the system to generate a sequence of weekly price estimates rather than restricting the output to a single prediction. This functionality is relevant to agricultural decision support because users may require an overview of expected price movements across several future weeks.

The six-week Anuradhapura forecast generated from 31 August 2026 showed relatively stable predicted values throughout the forecast period. This indicates that the implemented model produced a stable sequence of estimates for that particular district and forecasting scenario. Nevertheless, stability in predicted values should not automatically be interpreted as evidence of forecasting accuracy. Without comparison against actual observations over an appropriate evaluation period, it is not possible to determine whether the predicted stability reflects genuine market behavior or limitations in the model's response to changing conditions.

Consequently, the forecasting results support the operational feasibility of the developed approach but do not independently establish its ability to outperform alternative forecasting methods or accurately predict future paddy prices under different market conditions. A more conclusive interpretation of predictive

performance requires suitable held-out data, established error metrics, and comparisons with appropriate baseline models.

Interpretation of Explainability Results

The integration of SHAP-based feature contributions with dedicated XAI modules represents an important aspect of the developed component. Rather than presenting a numerical estimate in isolation, the system provides additional information about the factors contributing to the model output, the expected price trend, the application's confidence category, and the associated market outlook and recommendations.

These outputs can potentially make machine learning predictions easier to examine by providing information that helps users understand how the system has arrived at an estimate. The separation between prediction generation and explanation processing also supports a clearer system architecture: the XGBoost model produces the numerical prediction, while the XAI modules process supporting evidence for interpretation.

However, the existence of explanation outputs does not establish that those explanations are complete, faithful, or understandable to intended users. SHAP values describe the contribution of features to a model output under the selected explanation method; they do not, by themselves, demonstrate that those features causally determine paddy prices. Similarly, the application's confidence categories should be interpreted as system-defined indicators based on available explanation evidence rather than calibrated probabilities of prediction correctness.

Therefore, the findings support the successful implementation of an explainability mechanism, while the effectiveness of that mechanism as a means of improving user understanding requires further evaluation.

Interpretation of Natural-Language Explanation Results

The integration of the LLM explanation service extends the component by converting structured prediction evidence into English and simplified Sinhala explanations. This approach separates the generation of numerical predictions from the communication of their meaning, allowing the system to use a language model to explain existing evidence rather than relying on the language model to determine the predicted price.

The successful integration tests indicate that the structured evidence and natural-language explanation workflows operated as intended in the recorded test scenarios. The implementation of fallback behavior also provides a mechanism for handling failures in explanation generation separately from the underlying forecasting process.

These findings suggest that combining structured XAI evidence with natural-language generation is technically feasible within the developed architecture. The bilingual output is also consistent with the intention of making forecasting information accessible through the Digital Goviya application.

Nevertheless, technical success does not guarantee that every generated explanation accurately represents the underlying evidence or communicates uncertainty appropriately. The quality of the English and simplified Sinhala outputs, including their factual consistency, linguistic clarity, and suitability for agricultural contexts, has not been established through confirmed stakeholder evaluation. Further assessment is therefore necessary before concluding that this approach improves comprehension or supports better agricultural decisions.

Interpretation of Technical Performance and Reliability

The recorded performance measurements show that the prediction endpoint had the shortest average response time, followed by the forecasting endpoint and the LLM explanation endpoint. The higher response time observed for the LLM explanation endpoint indicates that the complete explanation-generation workflow required more time than the numerical prediction and recursive forecasting workflows in the tested environment.

This difference is consistent with the additional processing involved in generating natural-language explanations. However, the available measurements do not isolate the contribution of individual internal operations, so the observed difference should not be attributed exclusively to the language model.

The repeated-request tests also recorded successful responses for all 20 requests in one test and all 30 sequential requests in another. These results provide evidence that the prediction endpoint operated consistently during the recorded test sessions. In

addition, the correction of the unsupported-district validation issue demonstrates that systematic testing helped identify and resolve an error in request processing.

Despite these positive observations, the results are limited to the specific test environment and request patterns used. They do not establish behavior under concurrent user traffic, prolonged operation, network instability, or production deployment conditions. The technical results should therefore be interpreted as initial evidence of functional performance and repeated-request reliability rather than proof of production-scale robustness.

Interpretation of Integration Outcomes

The integration results demonstrate that the forecasting model, data and feature services, explanation modules, LLM explanation service, and FastAPI endpoints were connected into a modular backend workflow. This structure enables individual responsibilities to be separated while providing a common interface for requesting predictions and explanations.

The architecture is particularly relevant to the wider Digital Goviya ecosystem because the forecasting functionality can be accessed through APIs without requiring the mobile application to implement the machine learning and explainability processes directly. The separation between prediction and explanation also provides a basis for maintaining or modifying individual services independently.

However, successful backend integration should not be equated with complete validation of every application workflow. Although the recorded tests provide evidence of the operation of selected endpoints, the extent to which all functions have been validated end to end through the deployed mobile application has not been established by the available results.

Accordingly, the integration findings demonstrate that the main backend workflows are technically connected and testable, while complete application-level interoperability and deployment-scale behavior require separate evidence.

Overall Interpretation in Relation to the Research Aim

The overall findings indicate that the research aim of developing a machine learning-based paddy price forecasting component with integrated explainability and natural-language interpretation has been addressed at the software implementation and technical integration levels.

The component combines district-specific forecasting, recursive weekly prediction, SHAP-based feature contributions, structured XAI interpretations, and English and simplified Sinhala explanations within a common API-based architecture. The recorded functional, integration, performance, and repeated-request tests provide evidence supporting the operational feasibility of this design.

At the same time, the interpretation of the findings must remain within the boundaries of the available evidence. The successful execution of forecasts does not establish predictive accuracy; implemented explanations do not establish improved user understanding; and successful API responses do not demonstrate better decision-making among farmers or other stakeholders.

Three areas therefore remain important for further validation:

1. Predictive effectiveness: Evaluate forecasts against suitable held-out observations and compare their performance with appropriate baseline methods.
2. Explanation quality and usability: Assess the faithfulness, clarity, language quality, and practical interpretation of generated explanations.
3. Real-world applicability: Examine stakeholder perceptions and evaluate whether the information provided is relevant to agricultural planning and market-related decisions.

7.2 Comparison with literature baselines and existing solutions

Comparing the developed paddy price forecasting component with existing forecasting methods and agricultural decision-support solutions helps establish its position within the broader research field. The comparison considers not only the forecasting techniques used but also the interpretation of model outputs, the communication of

explanations, and the suitability of the resulting information for the Sri Lankan paddy supply chain.

An important distinction must be maintained between a comparison of methodological capabilities and an empirical comparison of predictive performance. The developed component uses XGBoost, recursive weekly forecasting, SHAP-based explanations, and an LLM-based explanation service. Although these technologies can be compared conceptually with methods reported in the literature, the available experimental results do not establish that the developed model achieves lower forecasting errors than ARIMA, Random Forest, LSTM, Transformer, or other baseline models. Such a conclusion would require comparable experiments using the same dataset, forecast horizons, and evaluation procedures.

Comparison with Existing Agricultural Decision-Support Solutions

The main distinction between the developed component and a prediction-only service is its combination of forecasting, interpretation, and communication.

A numerical forecasting service can return a predicted price, leaving users to interpret the value independently. In the developed component, the prediction can be accompanied by feature contributions, trend classifications, risk categories, recommendations, and a natural-language explanation. This arrangement is intended to provide additional context around the numerical estimate.

The English and simplified Sinhala outputs are also relevant to the intended Sri Lankan context. They provide a mechanism for presenting model-related information in two languages rather than exposing only technical features or raw model output. Nevertheless, the actual accessibility benefits remain a design intention until they are assessed with intended users.

The available evidence does not establish a comprehensive comparison against commercial agricultural applications, government information systems, or existing Sri Lankan paddy-market platforms. It would therefore be inappropriate to claim that the developed component offers more features or better outcomes than every existing solution.

Its defensible distinction is narrower: the implemented component integrates district-specific paddy price forecasting, SHAP-based evidence, rule-based interpretation modules, and bilingual LLM explanation generation within a single API-based backend workflow.

Implications for the Research Contribution

The comparison with existing methods suggests that the research contribution should be understood primarily as an integrated system design and implementation contribution, rather than as a demonstrated advancement in forecasting accuracy.

Three aspects are particularly relevant.

1. Context-specific forecasting: The component applies XGBoost and recursive weekly forecasting to the selected Sri Lankan paddy-price dataset and supported districts.
2. Integrated explanation: It combines prediction-level SHAP evidence with additional interpretation modules for trends, confidence categories, risk, and recommendations.
3. Accessible communication: It integrates an LLM explanation service that presents structured evidence in English and simplified Sinhala through the API architecture.

These elements form a coherent technical workflow. Nevertheless, their individual and combined value should be assessed through additional experiments, including comparisons against statistical and machine learning baselines, evaluation of explanation quality, and feedback from intended users.

Limitations of the Comparison

Several limitations affect the strength of the comparison.

First, the developed component has not been evaluated through a documented head-to-head experiment against ARIMA, Random Forest, LSTM, SARIMAX, or Transformer baselines using the same train-test split and forecast horizon. Published accuracy results cannot substitute for this missing experiment.

Second, the existing literature includes studies using different geographical regions, price series, observation periods, input variables, and forecasting objectives. Consequently, their reported metrics cannot be treated as directly comparable with the developed component's results without examining these methodological differences.

Third, the current tests establish that the software produces forecasts and explanations, but they do not demonstrate that the predictions are more accurate or that the explanations improve decision-making. The absence of confirmed stakeholder evaluation also limits conclusions about the practical advantages of the bilingual explanation functionality.

Finally, the available comparison is based on selected published forecasting methods and established explanation techniques, rather than a systematic audit of every existing agricultural decision-support product. Claims about the uniqueness of the complete system should therefore remain appropriately qualified.

7.3 Research contribution

This research contributes to the development of an explainable paddy price forecasting component designed for the Sri Lankan agricultural context. The primary contribution is the integration of machine learning-based price prediction, SHAP-based explainability, structured XAI interpretation, and large language model (LLM)-generated explanations within a unified backend service. The component is designed to provide not only numerical price estimates but also supporting information that helps users interpret forecast outputs within the Digital Goviya ecosystem.

The contribution should be understood in terms of the implemented system architecture, integration of complementary techniques, and application-oriented design. Although the recorded software tests demonstrate functional operation and technical feasibility, predictive superiority over existing models and improvements in farmer decision-making have not yet been empirically established.

Technical Contribution

The first contribution is the implementation of a modular forecasting backend that integrates a trained XGBoost model with historical data processing, feature generation, prediction services, and recursive multi-week forecasting.

The recursive forecasting mechanism enables the component to generate consecutive weekly estimates by incorporating each prediction into the data used for the subsequent forecasting step. This extends the service beyond single-step prediction and supports the generation of a multi-week price outlook for supported districts.

The use of FastAPI provides a standardized interface through which forecasting functionality can be accessed by the Digital Goviya mobile application and other authorized clients. The separation of data preparation, prediction, forecasting, and explanation services also supports modular development, testing, and maintenance.

The contribution lies in the implementation and integration of these components for the selected Sri Lankan paddy-price forecasting context. It does not imply that XGBoost or recursive forecasting is a newly invented algorithm.

Contribution to Explainable Price Forecasting

The second contribution is the integration of prediction-level explanation evidence with additional interpretation modules. The component uses SHAP to represent feature contributions and processes this information through modules responsible for trend classification, feature importance, confidence categorization, risk assessment, market outlook, and recommendations.

This design provides a structured explanation workflow around the numerical forecasting output. Instead of returning only a predicted price, the service can provide additional information describing the model output and its interpretation.

The approach also separates the prediction mechanism from the explanation mechanism. The XGBoost model remains responsible for generating the numerical estimate, while the XAI modules interpret the associated evidence.

This separation is an important architectural characteristic because it allows the explanation workflow to be developed and evaluated independently of the underlying forecasting model. Nevertheless, SHAP contributions are explanations of model behavior rather than proof of causal relationships, and the application's confidence categories should not be interpreted as calibrated probabilities of forecast correctness.

Contribution to Natural-Language and Bilingual Interpretation

The third contribution is the integration of an LLM-based explanation service that transforms structured forecasting evidence into natural-language explanations in English and simplified Sinhala.

The service receives information such as the predicted price, previous price, trend, confidence category, risk category, market outlook, recommendations, and important feature contributions. It then generates an explanation based on this supplied evidence rather than independently determining or modifying the numerical prediction.

This architecture connects model-level evidence with a user-oriented communication layer. It provides a mechanism for presenting technical forecasting information without requiring every user to interpret feature contributions or understand the internal operation of the XGBoost model.

The inclusion of simplified Sinhala is relevant to the intended Sri Lankan context, while English provides an additional language option. However, the actual improvement in accessibility, comprehension, or trust must be established through language-quality assessment and stakeholder evaluation; these benefits cannot be assumed from implementation alone.

Contribution to System Integration and Practical Application

The fourth contribution is the integration of forecasting, explainability, and natural-language generation behind a common REST API interface.

The component exposes separate endpoints for prediction, multi-week forecasting, structured explanation evidence, and LLM-generated explanations. This modular interface allows application clients to access the required functionality without implementing the machine learning and explanation processes independently.

The integration also incorporates request validation and centralized error handling. For example, the unsupported-district validation issue identified during testing was corrected so that district validity is checked before historical feature generation. The corrected behavior was subsequently retested.

These implementation outcomes demonstrate how the proposed component can function as a technical service within the wider Digital Goviya ecosystem. They

provide evidence of functional integration, although they do not independently establish complete end-to-end validation of every mobile application workflow.

Contribution to Empirical Software Evaluation

A further contribution is the evaluation of the implemented component through functional, unit, integration, performance, and repeated-request testing.

The recorded performance tests measured average response times of 53.3 ms for /predict, 105.7 ms for /forecast, and 371.0 ms for /explanation/llm. Separate sequential reliability tests recorded 20 successful responses from 20 prediction requests and 30 successful responses from 30 repeated prediction requests.

These results provide an initial empirical assessment of the operational behavior of the developed software under the tested conditions. They also help identify areas requiring further evaluation, including concurrent request handling, production-scale reliability, forecasting accuracy, and the quality of generated explanations.

The contribution is therefore an evidence-based technical evaluation of the implemented component, rather than a claim that the model has achieved superior predictive performance.

7.4 Product contribution

The product contribution of this research is the development of an explainable paddy price forecasting component designed for integration into the Digital Goviya ecosystem. The component combines machine learning-based price estimation, multi-week forecasting, prediction explanations, market-oriented interpretation, and bilingual natural-language communication within a single backend service.

Unlike a prediction-only implementation that returns numerical estimates, the developed component provides additional information to help users interpret forecasting outputs. Its product contribution therefore lies in the integration of forecasting and explanation capabilities into an application-oriented service designed for the Sri Lankan paddy supply chain.

Integrated Forecasting Service

The developed product provides district-specific paddy price estimates using a trained XGBoost model and supports recursive weekly forecasting. Through the FastAPI backend, the forecasting functionality is exposed through RESTful endpoints that can be accessed by compatible client applications.

The multi-week forecasting capability allows the system to generate a sequence of future price estimates, providing a broader view of expected price movements than a single-step prediction. This functionality is intended to support users who require price information across multiple weeks.

The implemented forecasting service establishes the core predictive functionality of the product. However, its practical value depends on predictive accuracy, data quality, and the relevance of the forecasting horizon to the user's needs.

Explainable Forecasting Outputs

A key product feature is the integration of explanation functionality with the numerical forecasting service. The component uses SHAP-based feature contributions alongside interpretation modules that generate trend classifications, feature-importance information, confidence categories, risk assessments, market outlooks, and recommendations.

These outputs provide additional context around the predicted price and are intended to make the information more informative for application users.

The product maintains a separation between numerical prediction and interpretation. The XGBoost model generates the prediction, while the explanation modules process supporting evidence. This design helps prevent the explanatory layer from being treated as the source of the numerical forecast.

The explanations remain decision-support information rather than guarantees of future market behavior. Their clarity and usefulness for actual users require further evaluation.

Bilingual Natural-Language Explanation

The product integrates an LLM-based explanation service that converts structured forecasting evidence into natural-language explanations in English and simplified Sinhala.

This feature is intended to reduce the need for users to interpret technical outputs independently. Instead of receiving only a predicted price and numerical feature contributions, users can receive a text-based explanation describing the forecast and its associated market interpretation.

The bilingual capability is particularly relevant to the intended Sri Lankan application context. Nevertheless, the implementation of English and simplified Sinhala outputs does not itself establish that every explanation is linguistically accurate or understandable to all users. These qualities require language review and user-centered evaluation.

Modular Architecture and Application Integration

The component is implemented as a modular FastAPI backend, with dedicated services for data loading, feature preparation, model loading, prediction, forecasting, explanation generation, and LLM integration.

This architecture allows the forecasting functionality to be accessed through a common API interface without requiring the client application to implement the internal machine learning processes. It also supports the independent testing and maintenance of individual modules.

The principal product endpoints include:

- /predict — generates a prediction and associated explanation.
- /forecast — generates recursive multi-week forecasts.
- /explanation/evidence — provides structured explanation evidence.
- /explanation/llm — generates natural-language explanations.
- /prediction/explanation — provides combined prediction explanation functionality.

This API-based design provides an integration foundation for the React Native Digital Goviya application. The extent of complete end-to-end integration should be assessed separately from backend endpoint testing.

Product Reliability and Operational Feedback

The development process incorporated request validation, centralized exception handling, and fallback behavior for LLM explanation failures. These features improve the product's ability to handle selected invalid requests and explanation-generation failures.

Testing also identified an issue in which an unsupported district was incorrectly reported as having insufficient historical data. The validation sequence was corrected, and the updated behavior was retested.

The recorded performance tests further demonstrated successful responses for the tested prediction, forecasting, and explanation requests. These results provide initial evidence of operational feasibility within the test environment, although they do not establish production-scale reliability or concurrent-user performance.

7.5 Limitations and threats to validity

Limitations and Threats to Validity

Although the developed Sri Lankan paddy price forecasting component demonstrates the technical feasibility of integrating machine learning, explainable artificial intelligence (XAI), and large language model (LLM)-based explanations, several limitations affect the interpretation and generalizability of the research findings.

These limitations relate to the available data, forecasting evaluation, explanation quality, software testing conditions, and the absence of confirmed stakeholder validation. Identifying these limitations is important to distinguish what has been demonstrated through implementation and testing from what remains to be established through further empirical research.

Data Limitations

The forecasting component relies on the available historical paddy-price dataset and its associated features. The representativeness, completeness, and consistency of these

data influence the information available to the trained model and, consequently, the predictions it produces.

The broader research dataset includes approximately ten years of historical paddy-price information, with weekly price observations and weather information. However, the available project records indicate gaps in some supplementary variables, including weather observations after 2023 and missing rainfall, fuel-price, and fertilizer-price information.

These gaps limit the range of external factors that can be represented in the forecasting process. Agricultural prices can be influenced by weather conditions, production levels, seasonal patterns, transportation costs, government policies, and changes in market demand. If relevant variables are unavailable or incompletely recorded, their influence may not be adequately represented in the model inputs.

Furthermore, the forecasting component currently supports selected districts and a selected paddy-price forecasting scope. Its findings should therefore not automatically be generalized to every district, paddy variety, or agricultural commodity in Sri Lanka.

Limitations of Forecasting Evaluation

A major limitation is that the available results establish the operational execution of the forecasting service but do not establish its predictive accuracy through a documented held-out evaluation.

The system successfully generated a six-week forecast for Anuradhapura and passed the recorded forecasting integration tests. However, successful generation of predictions does not demonstrate that those predictions closely match actual market prices.

Without documented evaluation metrics such as mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE), the magnitude of forecasting errors cannot be established from the available evidence. Similarly, without controlled comparisons against suitable statistical and machine learning baselines, the relative performance of XGBoost remains unknown.

The recursive forecasting approach introduces an additional consideration. Each generated prediction is incorporated into the data used for subsequent forecast steps. Consequently, errors in earlier predictions may influence later estimates. The extent of this effect has not been quantified in the recorded results.

Therefore, the current findings support the implementation and functional verification of the forecasting workflow, but not claims of superior forecasting accuracy or reliable performance across different forecast horizons.

Limitations of Explainability and LLM-Generated Explanations

The component combines SHAP-based feature contributions with additional XAI interpretation modules and LLM-generated explanations. Although the recorded tests support the implementation of these functions, several limitations remain concerning the reliability and interpretation of their outputs.

First, SHAP explanations describe how features contribute to a model's predictions under the selected explanation method. They do not independently establish causal relationships between individual factors and paddy-price movements. Users may misinterpret feature contributions if the explanations are presented without appropriate context.

Second, the confidence categories generated by the application are based on available explanation evidence rather than a calibrated statistical probability of prediction correctness. Consequently, these categories should not be treated as formal measures of the likelihood that a forecast will be accurate.

Third, LLM-generated explanations may introduce ambiguity, omit relevant qualifications, or express information in ways that do not perfectly reflect the supplied evidence. Although the service is designed to generate explanations from structured inputs and includes fallback behavior, successful API execution does not establish the factual consistency of every generated response.

Finally, the English and simplified Sinhala explanations have not been evaluated through documented stakeholder assessments or a formal explanation-quality benchmark. Their clarity, linguistic accuracy, faithfulness to the structured evidence,

and suitability for different levels of agricultural and technical knowledge therefore remain unverified.

Software Testing and Performance Limitations

The component underwent unit, integration, functional, performance, and repeated-request testing. These tests provide useful evidence about the operation of the implemented software, but their scope limits the conclusions that can be drawn about system reliability.

The performance evaluation used ten requests for each of the prediction, forecasting, and LLM explanation endpoints. The recorded average response times were 53.3 ms, 105.7 ms, and 371.0 ms, respectively. These measurements describe the tested environment and request conditions.

The sequential reliability tests recorded 20 successful responses from 20 requests and 30 successful responses from 30 requests. However, the latter test was performed sequentially rather than concurrently. The results therefore do not establish the system's ability to handle many simultaneous users.

Other limitations include the absence of a completed health-check performance evaluation and the lack of documented production-scale testing under varying network conditions, extended operation, and concurrent workloads. In addition, the recorded regression-test results after the unsupported-district correction were not confirmed in the available project evidence.

Accordingly, the results should be interpreted as evidence of successful execution within selected test scenarios rather than proof of comprehensive reliability or production readiness.

Limitations of Stakeholder Validation

The research intends to support the interpretation of paddy-price information through district-specific forecasts, explanatory evidence, and bilingual natural-language output. Nevertheless, the available evidence does not confirm the completion of interviews, questionnaires, usability tests, or other formal evaluations involving farmers, traders, agricultural officers, or other intended stakeholders.

As a result, several important questions remain unanswered:

- Whether intended users understand the forecast values and trend classifications.
- Whether the explanations communicate the relevant model evidence clearly.
- Whether simplified Sinhala explanations are appropriate for the intended audience.
- Whether users interpret confidence categories and recommendations responsibly.
- Whether the information is useful for agricultural planning or market-related decisions.

The absence of confirmed stakeholder feedback limits conclusions about the practical effectiveness and acceptability of the product. Although its features are intended to improve the accessibility of forecasting information, such benefits cannot be claimed as demonstrated outcomes without appropriate user-centered evidence.

7.6 Implications

The findings of this research have implications for the development of explainable machine learning applications, agricultural decision-support systems, and digital services intended for the Sri Lankan paddy supply chain. By integrating paddy price forecasting, SHAP-based explainability, structured interpretation modules, and bilingual natural-language explanations, the developed component provides a technical foundation for presenting forecasting information in a more contextualized form.

However, these implications should be interpreted in accordance with the available evidence. The research demonstrates technical implementation and selected software testing outcomes; the predictive effectiveness and practical benefits of the component require further empirical validation.

Technical Implications

The research demonstrates the feasibility of integrating multiple machine learning and explanation-related services within a modular backend architecture. The separation of data preparation, prediction, forecasting, explainability, and natural-language generation allows each responsibility to be developed and tested independently while remaining accessible through a common REST API interface.

This architecture has several technical implications. First, it provides a reusable structure for integrating forecasting functionality into agricultural applications without requiring the client application to implement the underlying machine learning processes. Second, it supports the maintenance and extension of individual services without necessarily redesigning the entire component. Third, it establishes a foundation for incorporating additional forecasting features or explanation mechanisms in future iterations.

The implementation also demonstrates the importance of separating numerical prediction from its interpretation. By keeping the XGBoost prediction process distinct from the XAI and LLM explanation layers, the system preserves a clear distinction between the predicted value and the narrative used to communicate its meaning.

Nevertheless, the architecture's scalability, maintainability over extended periods, and suitability for production workloads require further evaluation beyond the recorded tests.

Implications for Explainable Artificial Intelligence

The research illustrates how SHAP-based feature contributions can be combined with additional interpretation modules to produce structured explanations of agricultural price predictions. Rather than presenting only numerical outputs, the component provides trend classifications, feature-importance information, confidence categories, risk assessments, market outlooks, and recommendations.

This approach has implications for the design of explainable machine learning applications because it combines model-level evidence with application-level interpretation. It provides a framework through which technical information can be organized into outputs that are potentially easier for non-specialist users to examine.

The integration also highlights the need to communicate explanation limitations responsibly. Feature contributions should not be presented as proof of causality, and application-defined confidence categories should not be interpreted as calibrated probabilities of forecast correctness.

Consequently, the research supports the feasibility of integrating explainability into an agricultural forecasting workflow, while emphasizing that the quality, faithfulness, and usefulness of explanations must be evaluated independently.

Implications for Sri Lankan Agriculture

Access to information about potential future paddy prices may be relevant to farmers, traders, warehouse operators, and agricultural advisors who consider market conditions when planning agricultural activities.

The developed component is designed to provide district-specific price estimates and multi-week forecasts, together with explanations and market-oriented interpretations. These outputs could potentially contribute to informed consideration of selling periods, storage decisions, and market conditions when used alongside other relevant information.

The intended value is not that the forecasting component can determine the best agricultural decision independently. Rather, it can provide an additional source of information that stakeholders may consider together with actual market prices, production conditions, storage costs, government policies, and their individual circumstances.

The practical implications remain conditional because the current research has not established forecasting superiority or demonstrated improvements in agricultural decisions through stakeholder evaluation. Further validation using observed market data and feedback from intended users is necessary before making stronger claims about agricultural benefits.

Implications for Language Accessibility and Human–AI Interaction

The integration of English and simplified Sinhala explanations has implications for presenting machine learning outputs in a context where users may have different language preferences and levels of technical knowledge.

By transforming structured prediction evidence into natural-language explanations, the component provides an alternative to displaying numerical forecasts and technical feature contributions alone. This may help users engage with the information without requiring detailed knowledge of XGBoost or SHAP.

The design also demonstrates a potential role for LLMs in communicating machine learning evidence. The language model is used to explain information generated by the forecasting and interpretation services rather than independently determining the numerical prediction.

However, language accessibility depends on more than providing multiple language outputs. Explanation accuracy, linguistic quality, appropriate uncertainty communication, and user comprehension must also be assessed. Therefore, the bilingual capability represents an implemented product feature and a potential accessibility benefit, rather than a confirmed improvement in user understanding.

Implications for Digital Agricultural Platforms

The use of REST APIs provides a technical basis for integrating the forecasting component into the wider Digital Goviya ecosystem. This approach allows the forecasting and explanation functionality to be accessed through application interfaces while keeping the machine learning processes within the backend.

For digital agricultural platforms, this architecture illustrates how predictive analytics can be combined with interpretive services as part of a broader application. The same architectural principle may be relevant to other agricultural information services that require numerical predictions and explanatory outputs.

However, applying this architecture to other agricultural domains would require appropriate datasets, domain-specific model development, and separate evaluation. The current results should not be interpreted as evidence that the developed model can be transferred directly to other commodities or agricultural contexts.

Overall Implications

Overall, this research demonstrates the technical feasibility of combining machine learning-based paddy price forecasting, explainable AI, and bilingual natural-language interpretation within a modular backend designed for the Digital Goviya ecosystem.

Its main implication is that agricultural forecasting services can be architected to provide both numerical estimates and structured explanations through a common application interface. This creates a foundation for further research into the communication and interpretation of agricultural forecasting information.

At the same time, the distinction between technical feasibility and demonstrated practical benefit remains essential. Predictive accuracy, explanation quality, stakeholder acceptance, and improvements in agricultural decision-making require additional empirical evidence.

The research therefore contributes a practical technical foundation for future work on explainable digital agricultural services in Sri Lanka, while identifying the validation needed to establish their effectiveness in real-world use.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research focused on the development of a machine learning-based paddy price forecasting component for the Sri Lankan agricultural context, integrating explainable artificial intelligence (XAI) and large language model (LLM)-based natural-language explanations. The component was developed as part of the wider Digital Goviya ecosystem, with the aim of combining district-specific price estimation, multi-week forecasting, prediction interpretation, and application-oriented communication within a unified backend service.

The developed solution integrates a trained XGBoost model with historical data processing, feature generation, and recursive weekly forecasting. Through the implemented FastAPI backend, the component provides endpoints for generating price predictions, producing multi-week forecasts, and accessing associated explanation services. The recursive forecasting mechanism enables the generation of consecutive weekly estimates, extending the functionality beyond single-step prediction.

A key outcome of the research is the integration of SHAP-based feature contributions with dedicated XAI interpretation modules. These modules provide trend classifications, feature-importance information, application-defined confidence categories, risk assessments, market outlooks, and recommendations. By combining these outputs with numerical predictions, the component provides a structured approach to presenting additional context around forecasting results. However, the generated explanations describe model behavior and associated interpretations; they do not independently establish causal relationships or guarantee future market outcomes.

The research also integrated an LLM-based explanation service to transform structured prediction evidence into natural-language explanations in English and simplified Sinhala. This approach separates numerical prediction from explanation generation, allowing the language model to communicate existing forecasting evidence rather than independently determine or modify the predicted price. The recorded integration tests confirmed the operation of the structured evidence and LLM explanation workflows,

demonstrating the technical feasibility of incorporating bilingual explanation generation into the forecasting architecture.

The developed component was evaluated through unit testing, integration testing, functional testing, response-time measurements, and repeated-request testing. The recorded performance tests produced average response times of 53.3 milliseconds for the prediction endpoint, 105.7 milliseconds for the forecasting endpoint, and 371.0 milliseconds for the LLM explanation endpoint. Separate sequential reliability tests recorded successful responses for all 20 requests in one test and all 30 requests in another. These results provide initial evidence of functional operation and response performance under the tested conditions. They do not establish production-scale reliability or concurrent-user capacity.

The research also identified and corrected an error in which requests for unsupported districts were incorrectly reported as insufficient historical data. The correction and subsequent retesting demonstrated the importance of request validation and systematic software testing in improving the reliability of the component's behavior.

Despite these implementation achievements, several limitations remain. The available results do not establish forecasting accuracy through documented held-out evaluation metrics, demonstrate superiority over statistical or alternative machine learning baselines, or confirm improvements in user comprehension and agricultural decision-making. Furthermore, the quality of the generated explanations and the practical usefulness of the component have not been established through confirmed stakeholder validation.

In conclusion, the research demonstrates the technical feasibility of integrating machine learning-based paddy price forecasting, explainable AI, and bilingual natural-language explanation generation within a modular backend for the Digital Goviya ecosystem. Its principal contribution is the implementation of an integrated forecasting and explanation service designed for the Sri Lankan paddy supply chain context. The findings provide a foundation for further investigation into predictive effectiveness, explanation quality, system scalability, and practical applicability. Establishing these

benefits through controlled experiments and stakeholder evaluation remains an essential direction for future research.

8.2 Achievement of objectives

This section evaluates the extent to which the research objectives established for the Sri Lankan paddy price forecasting component were achieved. The assessment is based on the implemented functionality, recorded software testing results, and integration outcomes discussed in the preceding chapters. It distinguishes objectives supported by implementation and testing evidence from those requiring further empirical validation.

Achievement of the Research Objectives

Research objective	**Implementation and evidence**	**Achievement status**
Develop a machine learning-based paddy price forecasting component for selected Sri Lankan districts	Integrated a trained XGBoost model with data processing, feature generation, and prediction services. Prediction functionality was tested successfully.	Achieved at the implementation level
Implement multi-week paddy price forecasting	Developed recursive weekly forecasting and generated a six-week forecast for Anuradhapura. Forecast integration tests passed.	Achieved at the implementation level
Integrate explainable AI techniques to interpret predictions	Integrated SHAP-based feature contributions with trend, feature-importance, confidence, risk, and recommendation modules. Recorded unit and integration tests support implementation.	Achieved at the implementation level

Generate human-readable explanations of forecasting results	Integrated an LLM explanation service that uses structured prediction evidence to generate English and simplified Sinhala explanations. The recorded integration test passed.	Achieved at the implementation level
Integrate forecasting and explanation functionality through a modular API	Implemented FastAPI endpoints for prediction, forecasting, structured evidence, and LLM explanations. Selected endpoints were tested.	Achieved for the tested backend workflows
Evaluate the component's technical operation and performance	Conducted functional, unit, integration, response-time, and sequential repeated-request tests.	Partially achieved; evaluation scope remains limited
Establish predictive effectiveness and practical stakeholder usefulness	The available results do not include documented held-out forecasting accuracy metrics, controlled baseline comparisons, or confirmed stakeholder evaluation.	Not conclusively established

Table 18 : Summary of Research Objective Achievement

Discussion of Objective Achievement

The first objective was addressed through the implementation of the XGBoost-based prediction service. The integration of data loading, feature preparation, and model inference enabled the component to generate district-specific price estimates for supported inputs. This establishes the operational implementation of the forecasting functionality, although predictive accuracy remains to be established through a documented held-out evaluation.

The second objective was addressed through the implementation of recursive weekly forecasting. The service generated consecutive estimates over a six-week period, demonstrating that the forecasting workflow can produce multi-week outputs.

However, the accuracy of these estimates over different forecast horizons has not been established through the available results.

The third objective was addressed by integrating SHAP-based feature contributions with additional XAI interpretation modules. The resulting architecture provides structured information about model behavior, predicted trends, confidence categories, risk, and recommendations. Recorded testing supports the implementation of these functions, but the quality and usefulness of the explanations have not been independently established.

The fourth objective was addressed through the integration of the LLM explanation service. The service uses structured forecasting evidence to generate explanations in English and simplified Sinhala. Successful integration testing supports the technical feasibility of this workflow. Nevertheless, the accuracy, linguistic quality, and user comprehension of the generated explanations require further evaluation.

The fifth objective was addressed through the modular FastAPI architecture. The implemented endpoints provide a common interface for accessing prediction, forecasting, and explanation functionality. The recorded tests support the operation of the selected backend workflows, although complete end-to-end validation of every mobile application workflow has not been established by the available evidence.

The technical evaluation objective was addressed through selected functional, unit, integration, performance, and repeated-request tests. These activities provided evidence of software functionality and response performance under the tested conditions. However, additional concurrent-load testing, production-environment evaluation, and comprehensive regression testing would be required to establish broader operational reliability.

Finally, predictive effectiveness and practical stakeholder usefulness remain areas requiring further investigation. The available evidence does not establish whether the XGBoost model outperforms alternative forecasting methods or whether the explanations improve users' understanding and agricultural decision-making. These outcomes require appropriate comparative experiments and direct stakeholder evaluation.

Overall Assessment

Overall, the research objectives concerning software implementation, recursive forecasting, explainability integration, natural-language explanation generation, and backend integration were addressed at the implementation level. Recorded software tests provide supporting evidence for the functional operation of the principal component workflows.

However, the objectives relating to predictive effectiveness, comparative model performance, and practical stakeholder impact cannot be considered conclusively established from the current evidence.

The research therefore achieved its principal technical aim of developing an integrated and explainable paddy price forecasting component for the Digital Goviya ecosystem. Further empirical evaluation is necessary to determine the extent to which this component improves forecasting performance, explanation quality, and practical agricultural decision support.

8.3 Individual research and product contributions

This research was conducted as an individual component within the wider Digital Goviya project, which aims to support the modernization of the Sri Lankan paddy supply chain through digital technologies. The individual research focused specifically on developing a machine learning-based paddy price forecasting component integrated with explainable artificial intelligence (XAI) and large language model (LLM)-based natural-language explanations.

The individual contribution covered the design, implementation, integration, testing, evaluation, and documentation of the forecasting component. The work was directed towards developing a functional backend service capable of generating district-specific price estimates, producing multi-week forecasts, explaining model outputs, and communicating the resulting information through an API-based architecture.

Individual Research Contributions

The primary research contribution was the investigation and implementation of an integrated approach to explainable paddy price forecasting for the Sri Lankan agricultural context.

The individual research contributions included the following:

1. Machine learning-based forecasting: Implemented the integration of the trained XGBoost model with the data-loading, feature-generation, and prediction services to generate price estimates for supported districts and dates.

2. Recursive multi-week forecasting: Developed the forecasting workflow to generate consecutive weekly price estimates by incorporating earlier predictions into subsequent forecasting steps.

3. Explainable artificial intelligence: Integrated SHAP-based feature contributions to provide evidence about how input features influence model predictions.

4. Structured XAI interpretation: Implemented interpretation modules for trend classification, feature importance, confidence categorization, market outlook, risk assessment, recommendations, and summaries.

5. Evidence-grounded natural-language explanations: Integrated an LLM explanation service that converts structured prediction evidence into human-readable explanations in English and simplified Sinhala, while keeping numerical prediction generation separate from language generation.

6. Research evaluation: Conducted software testing and recorded performance and repeated-request measurements to evaluate the functional operation of the implemented component.

7. Research documentation: Documented the component architecture, implementation approach, testing procedures, results, limitations, and implications to establish a traceable account of the research and its outcomes.

These contributions connect the research objectives with the practical implementation of an explainable forecasting service. However, the research does not claim that the selected forecasting model outperforms alternative algorithms or that the generated

explanations improve farmer decision-making, as these outcomes have not been established by the available empirical evidence.

Integration with the Wider Digital Goviya Project

The developed forecasting component was designed to operate as a backend service within the wider Digital Goviya ecosystem. Its API-based architecture allows compatible client applications, including the React Native mobile application, to request price predictions, multi-week forecasts, and associated explanations without implementing the forecasting logic directly within the client.

The individual contribution focused on the forecasting and explainability backend, including its internal services, API endpoints, validation mechanisms, and testing. The wider ecosystem provides the application context in which these capabilities are intended to be consumed.

This distinction is important because the research contribution concerns the forecasting and explanation component rather than claiming sole responsibility for every module of the complete Digital Goviya platform.

8.4 Future directions

Although the developed paddy price forecasting component demonstrates the technical feasibility of integrating machine learning, explainable artificial intelligence (XAI), and large language model (LLM)-based explanations, several opportunities remain to improve its predictive performance, explanation quality, operational reliability, and practical applicability. Future development should focus on addressing the limitations identified in this research and extending the component to support more comprehensive agricultural decision-making within the Digital Goviya ecosystem.

Improving Forecasting Accuracy and Model Evaluation

A primary direction for future research is to conduct a comprehensive evaluation of the forecasting model using appropriate held-out observations and time-series validation techniques. Although the current implementation successfully generates district-specific predictions and recursive weekly forecasts, its predictive accuracy has not been established through documented held-out evaluation metrics.

Future experiments should compare XGBoost with suitable baseline methods, including ARIMA, Random Forest, LSTM, and other relevant forecasting approaches. Evaluation metrics such as mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE), where appropriate, could be used to quantify prediction errors.

Time-series-aware validation should be adopted to preserve the chronological order of observations and reduce the risk of information leakage. Such experiments would help determine whether XGBoost provides suitable performance for the selected paddy-price dataset and how its performance compares with alternative approaches across different forecast horizons and districts.

Expanding and Improving Data Quality

The predictive capability of the component may be improved by expanding the range, completeness, and consistency of the input data. The current research identified gaps in supplementary variables, including weather information after 2023 and missing rainfall, fuel-price, and fertilizer-price observations.

Future work could investigate the integration of more complete weather records, rainfall measurements, production statistics, fertilizer prices, transportation costs, and other relevant market indicators, subject to data availability and reliability.

Additional data-quality procedures could also be developed to identify missing observations, inconsistent records, and unusual price movements. The effects of different missing-data treatments should be evaluated rather than assuming that any single imputation method will improve forecasting performance.

Expanding the dataset across additional districts, paddy varieties, and observation periods could further support the investigation of geographical and temporal differences in price behavior. Any extension should be accompanied by appropriate validation to determine whether the model generalizes to the additional data.

Advancing Recursive Forecasting

The current forecasting service uses a recursive approach to generate consecutive weekly estimates. Although this mechanism supports multi-week forecasting, errors in earlier predictions may affect subsequent estimates.

Future research could compare recursive forecasting with direct multi-step forecasting and other suitable strategies. These experiments could evaluate whether alternative methods reduce error accumulation or improve performance over longer forecast horizons.

Forecasting performance should also be evaluated separately for different horizons, such as one-week, four-week, and longer-term estimates, where supported by the available data. This would help identify the periods for which the model provides the most reliable estimates.

In addition, future versions could investigate the generation of prediction intervals to communicate uncertainty around forecast values. Such intervals should be evaluated for appropriate coverage and calibration before being presented as reliable measures of uncertainty.

Enhancing Explainability and Confidence Assessment

The existing component integrates SHAP-based feature contributions with structured interpretation modules. Future research could evaluate the faithfulness, stability, and consistency of these explanations under different input conditions.

Comparative experiments could investigate alternative explanation methods, including LIME, to determine whether different techniques provide complementary insights for the selected forecasting model.

The application-defined confidence categories could also be improved by developing and validating a more rigorous uncertainty-assessment framework. Future work should distinguish model uncertainty, data limitations, and explanation evidence rather than treating them as interchangeable concepts.

Where sufficient evaluation data are available, calibrated uncertainty estimates could be investigated and compared with the existing confidence categories. This would help

ensure that the system communicates uncertainty more meaningfully without presenting confidence labels as guarantees of predictive correctness.

Improving LLM Explanation Quality and Language Accessibility

The LLM explanation service provides English and simplified Sinhala explanations based on structured prediction evidence. Future development could focus on evaluating and improving the factual consistency, linguistic quality, clarity, and completeness of these explanations.

A structured evaluation framework could compare generated explanations against the underlying numerical prediction and SHAP evidence. Expert review and carefully designed evaluation tasks could help identify cases where explanations omit important qualifications, misrepresent feature contributions, or communicate uncertainty inadequately.

Further work could also investigate improvements to Sinhala terminology, consistency of agricultural vocabulary, and the suitability of explanations for users with different levels of technical knowledge.

In addition, the explanation service could be evaluated against alternative prompt designs or constrained output-generation approaches. The objective would be to improve explanation reliability while preserving the separation between the forecasting model and the language-generation layer.

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GLOSSARY

The following glossary defines key technical terms, concepts, and abbreviations used in this research on explainable paddy price forecasting for the Sri Lankan agricultural context.

Term	Definition
Application Programming Interface (API)	An interface that enables software components to communicate and exchange data according to defined rules.
Artificial Intelligence (AI)	A field of computing concerned with developing systems capable of performing tasks that typically require human-like intelligence.
Confidence Category	An application-defined classification used to communicate an interpretation of available prediction explanation evidence. In this research, it is not a calibrated probability of forecast correctness.
Data Preprocessing	The process of preparing raw data for analysis or model training, including handling missing values, correcting inconsistencies, and transforming data into a suitable format.
Digital Goviya	The broader digital agricultural ecosystem within which the paddy price forecasting component is designed to operate.
District-Specific Forecasting	Generating paddy price estimates for a particular geographical district using the available historical data and model inputs.
Explainable Artificial Intelligence (XAI)	Techniques and methods designed to make the behavior and outputs of artificial intelligence models more understandable to users.

Feature	An input variable used by a machine learning model to generate a prediction.
Feature Importance	An indication of the relative contribution or significance of input features to a model or its predictions, depending on the method used.
FastAPI	A Python framework used to develop web APIs, including the REST endpoints implemented in this research.
Forecast Horizon	The period into the future covered by a forecast, such as the number of weeks for which paddy prices are estimated.
Large Language Model (LLM)	A language model capable of generating and interpreting text. In this research, it is used to communicate structured forecasting evidence through natural-language explanations.
Machine Learning (ML)	A field of artificial intelligence in which models learn patterns from data to perform tasks such as prediction and classification.
Mean Absolute Error (MAE)	A forecasting evaluation metric that measures the average absolute difference between predicted and actual values.
Mean Absolute Percentage Error (MAPE)	A metric that expresses the average absolute prediction error as a percentage of actual values. Its suitability depends on the presence of zero or near-zero actual values.
Model Inference	The process of using a trained machine learning model to generate predictions from input data.
Natural-Language Explanation	A text-based description intended to communicate prediction-related information in language that users can understand.

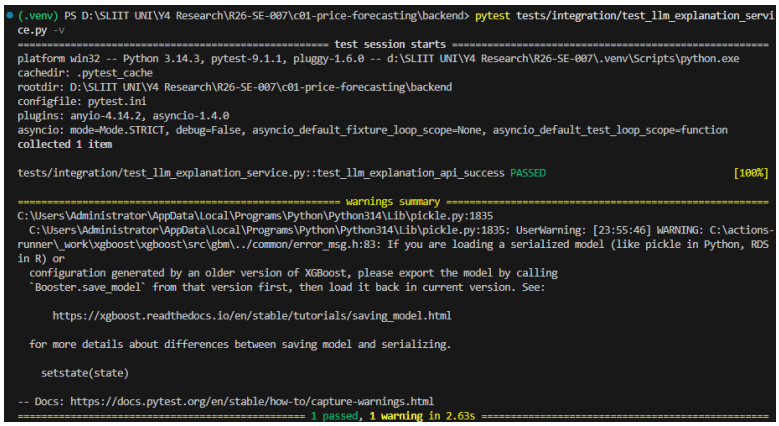
Paddy	Rice grain that has been harvested but retains its outer husk. Paddy prices are the subject of the forecasting component developed in this research.
Prediction	An estimate generated by a trained model for a specified input or observation.
Recursive Forecasting	A multi-step forecasting approach in which a prediction generated for one future step is incorporated into the input data used to generate a subsequent prediction.
Representational State Transfer (REST)	An architectural style commonly used to design web services that expose resources and operations through HTTP-based interfaces.
Root Mean Squared Error (RMSE)	A forecasting evaluation metric calculated as the square root of the mean squared difference between predicted and actual values. It gives greater weight to larger errors.
SHAP (SHapley Additive exPlanations)	An explainability method based on Shapley values that estimates the contribution of input features to individual model predictions.
Time-Series Data	Observations collected or recorded over time, such as weekly paddy prices.
Trend Classification	Categorizing expected price movement into predefined groups, such as increase, decrease, or stable.
XGBoost	An optimized gradient-boosting machine learning library used in this research to generate paddy price estimates.

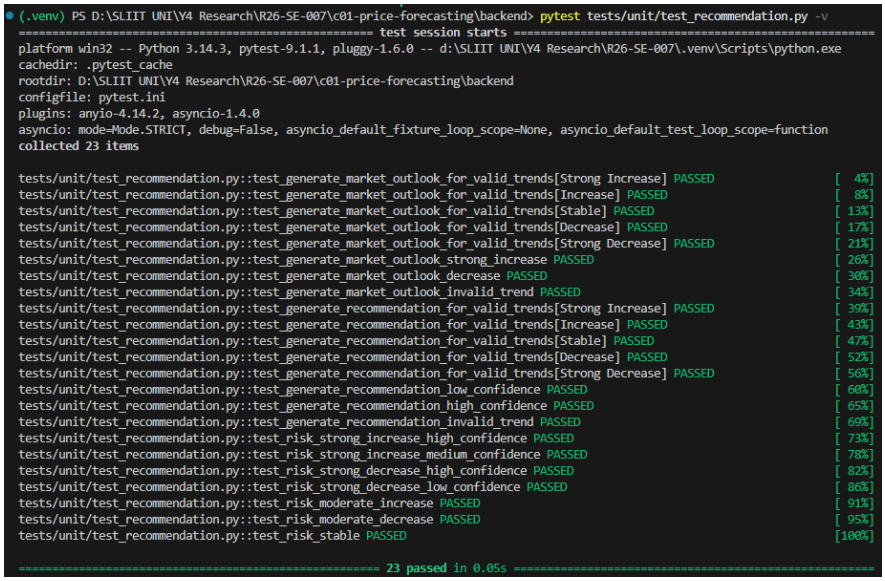
APPENDICES

Appendix A Research instruments or extended results

This appendix provides supplementary technical and evaluation materials supporting the development and assessment of the Sri Lankan paddy price forecasting component. It contains the API request structures used to interact with the forecasting services, supplementary forecasting outputs, recorded software testing results, performance measurements, and selected validation scenarios.

The purpose of this appendix is to provide additional evidence supporting the implementation and evaluation described in the main chapters without interrupting the discussion of the principal findings.



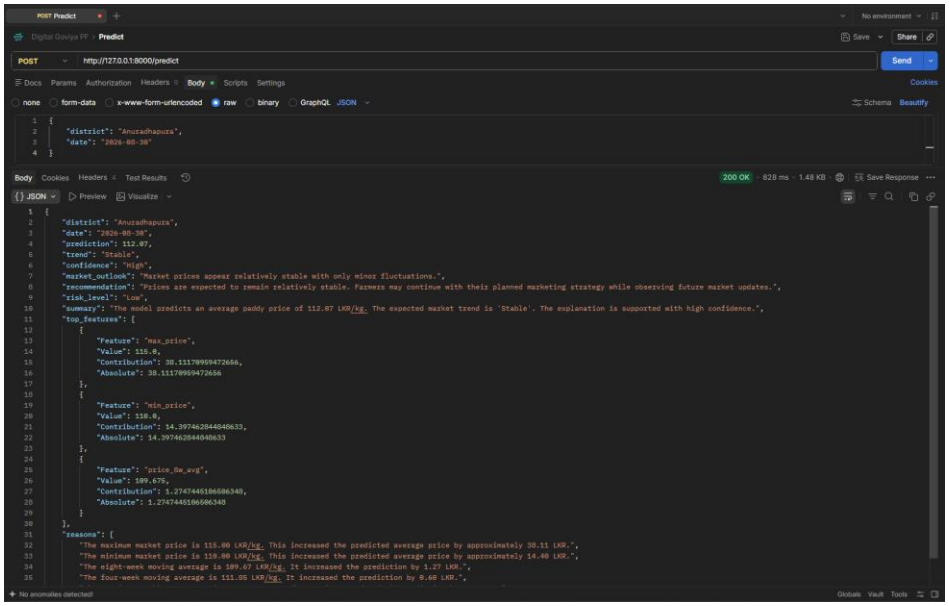


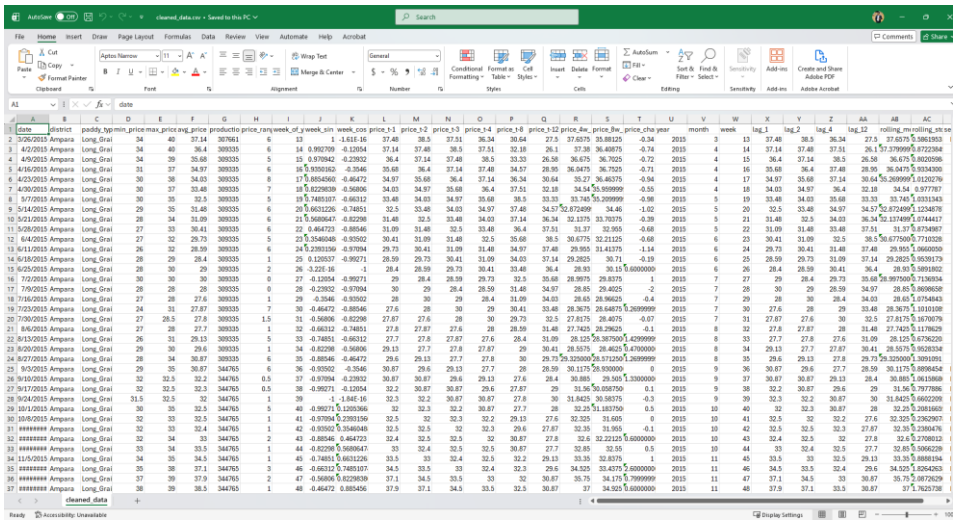
Price Trend

```
plt.figure(figsize=(12,5))
plt.plot(df['date'], df['avg_price'])
plt.title("Paddy Price Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Price")
plt.show()
```



Test	Test Type	Module	Test Description	Expected Result	Actual Result	Stat
UT-	Unit	Trend	Test strong increase	StrongIncrease	StrongIncrease	PASS
UT-	Unit	Trend	Test increase	Increase	Increase	PASS
UT-	Unit	Trend	Test stable	Stable	Stable	PASS
UT-	Unit	Trend	Test decrease	Decrease	Decrease	PASS
UT-	Unit	Trend	Test strong decrease	StrongDecrease	StrongDecrease	PASS
UT-	Unit	Trend	Test zero previous price	Stable	Stable	PASS
UT-	Unit	Confidence	High Confidence	High	High	PASS
UT-	Unit	Confidence	Medium Confidence	Medium	Medium	PASS
UT-	Unit	Confidence	Low Confidence	Low	Low	PASS
UT-	Unit	Confidence	Zero SHAP Values	Low	Low	PASS
UT-	Unit	Feature Importance	Top N features			PASS
UT-	Unit	Feature Importance	SHAP sorting			PASS
UT-	Unit	Feature Importance	Most important feature			PASS
UT-	Unit	Feature Importance	Feature value preserved			PASS
UT-	Unit	Feature Importance	SHAP contribution preserved			PASS
UT-	Unit	Dynamic Reasons	Top N reasons			PASS
UT-	Unit	Dynamic Reasons	String output			PASS
UT-	Unit	Dynamic Reasons	Non-empty reasons			PASS
UT-	Unit	Dynamic Reasons	Top N respected			PASS
UT-	Unit	Recommendation	Verify market outlook generation for Strong Increase, Increase, Stable, Decrease, and Strong Decrease	Appropriate market outlook is returned for each trend.		PASS
UT-	Unit	Recommendation	Verify that Strong Increase generates an up and market outlook.	Outlook should indicate an upward price movement.	Correct market outlook returned for all five trends.	PASS
UT-	Unit	Recommendation	Verify that Decrease generates a down and market outlook.	Outlook should indicate a downward price trend.	Up and price movement is as correctly returned.	PASS
UT-	Unit	Recommendation	Verify fallback behavior for an unsupported trend.	System should return "Market outlook unavailable."	Down and price trend was correctly returned.	PASS
UT-	Unit	Recommendation	Verify recommendation generation for all five supported trends.	Appropriate recommendation should be returned for each trend.	Correct fallback message was returned.	PASS
UT-	Unit	Recommendation	Verify that low confidence adds an appropriate caution to the recommendation.	Recommendation should mention low prediction confidence.	Correct recommendations returned for all tested trends.	PASS
UT-	Unit	Recommendation	Verify that high confidence does not add the 'low confidence warning.	Recommendation should not contain a low-confidence warning.	Low confidence warning was included.	PASS
UT-	Unit	Recommendation	Verify fallback behavior for an unsupported trend.	System should return "No recommendation available."	No low-confidence warning was included.	PASS
UT-	Unit	Recommendation	Verify risk assessment for strong increase with high confidence.	Risk level should be High.	Correct fallback message was returned.	PASS
UT-	Unit	Recommendation	Verify risk assessment for strong increase with medium confidence.	Risk level should be Medium.	High risk correctly returned.	PASS
UT-	Unit	Recommendation	Verify risk assessment for strong decrease with high confidence.	Risk level should be High.	Medium risk correctly returned.	PASS
UT-	Unit	Recommendation	Verify risk assessment for strong decrease with low confidence.	Risk level should be Medium.	High risk correctly returned.	PASS
UT-	Unit	Recommendation	Verify risk assessment for Increase trend.	Risk level should be Medium.	Medium risk correctly returned.	PASS
UT-	Unit	Recommendation	Verify risk assessment for Decrease trend.	Risk level should be Medium.	Medium risk correctly returned.	PASS
UT-	Unit	Recommendation	Verify risk assessment for Stable trend.	Risk level should be Low.	Low risk correctly returned.	PASS
UT-	Unit	Summary	Verify summary generation for a high-confidence prediction.	Summary should contain prediction price, LKR/kg, trend, and confidence.	All required information was correctly included.	PASS
UT-	Unit	Summary	Verify summary generation for a medium-confidence prediction.	Summary should contain prediction price, LKR/kg, trend, and confidence.	All required information was correctly included.	PASS
UT-	Unit	Summary	Verify summary generation for a low-confidence prediction.	Summary should contain prediction price, LKR/kg, trend, and confidence.	All required information was correctly included.	PASS
UT-	Unit	Summary	Verify that the predicted price is formatted to two decimal places.	Prediction should be displayed with two decimal places.	Prediction was correctly formatted to two decimal places.	PASS
UT-	Unit	Summary	Verify that the generated summary contains prediction, trend, and confidence information.	Summary should contain all required prediction information.	All required information was correctly included.	PASS
IT-001	Integration Test	/predict API	End-to-End Prediction and XAI Integration	API should return HTTP 200 with prediction, trend, confidence, top features, reasons, market outlook, recommendation, risk level, and summary	/predict successfully processed the request and returned all required prediction and XAI fields	PASS
IT-002	Integration Test	/forecast API	Verify that the forecast endpoint integrates request validation, feature generation, XGBoost prediction, and recursive multi-week forecasting.	API should return the requested number of weekly forecasts with week number, date, and predicted price.	API successfully generated 6 weekly forecasts with the required fields.	PASS
IT-003	Integration Test	/forecast API	Verify that the forecasting service works correctly for different supported districts.	API should successfully generate forecasts for the requested district.	API successfully generated forecasts for Polonnaruwa.	PASS
IT-004	Integration Test	/explain and /explain/stream API	Verify that the explanation endpoint integrates prediction generation, feature generation, SHAP analysis, trend classification, confidence estimation, risk assessment, market outlook, recommendation, and SHAP-based reasons.	API should return HTTP 200 with valid prediction evidence, trend, confidence, risk level, recommendations, top SHAP features, and explanation reasons.	API successfully generated the complete explanation evidence with all required fields and valid SHAP feature information.	PASS
IT-005	Integration Test	/explain and /explain/stream API	Verify integration of XAI evidence generation, LLM explanation service, LLM client, response validation, and API response generation.	API should successfully return structured LLM explanation containing headline, explanation, key factors, farmer summary, and generation source.	API successfully generated and validated the LLM explanation with all required fields.	PASS
ST-001	System Test	Normal Paddy Price Prediction	District: Anuradhapura, Date: 2026-08-30	System should generate a paddy price prediction with XAI-based trend, confidence, feature importance, market outlook, recommendation and risk information.	System successfully generated the prediction and complete XAI information without errors.	PASS
ST-002	System Test	Prediction for Different District	District: Polonnaruwa, Date: 2026-08-30	System should generate a valid prediction and explanation for a different supported district.	System successfully generated the prediction and XAI results for Polonnaruwa.	PASS
ST-003	System Test	8-Week Paddy Price Forecast	District: Anuradhapura, Start Date: 2026-08-30, Weeks: 6	System should generate eight sequential weekly price forecasts with corresponding dates and predicted prices.	System generated six weekly forecasts with sequential dates and predicted prices.	PASS
ST-004	System Test	LLM-Based Human-Friendly Explanation	District: Anuradhapura, Date: 2026-08-30	System should generate a human-readable explanation based on the model prediction and XAI evidence without changing the predicted result.	LLM generated the explanation successfully using the provided prediction evidence.	PASS
ST-005	System Test	Sinhala Human-Friendly Explanation	Anuradhapura, 2026-08-30	System should provide a simple, farmer-friendly Sinhala explanation while preserving the meaning and numerical information of the prediction.	Sinhala explanation was generated using simple, understandable terminology and preserved the prediction meaning.	PASS
ST-006	System Test	English Human-Friendly Explanation	Anuradhapura, 2026-08-30	System should generate a clear and understandable English explanation based only on the supplied model evidence.	English explanation was generated successfully and remained consistent with the model evidence.	PASS





This appendix provides supplementary evidence for the implementation and technical evaluation of the paddy price forecasting component. The included forecast example, API endpoint inventory, performance measurements, repeated-request results, and selected test outcomes support the technical findings presented in the main thesis.

The appendix does not establish forecasting accuracy against actual market prices, comparative superiority over alternative models, formal stakeholder acceptance, or improved agricultural decision-making. These outcomes require additional empirical evaluation.

Before submitting: Verify the example request against your actual Pydantic request schema, add the referenced screenshots where available, and ensure the final test counts reflect the latest confirmed regression-test results.

Appendix B Academic integrity and AI use disclosure

Academic Integrity Statement

This research was conducted with the intention of maintaining academic integrity, transparency, and responsible research practices. The research objectives, system design, implementation, testing, evaluation, and documentation were developed in accordance with the applicable academic requirements of the degree programme.

Technical findings and reported evaluation results are based on the implementation and testing evidence available for the paddy price forecasting component. Limitations in the available evidence have been acknowledged, and claims regarding forecasting

accuracy, stakeholder acceptance, and practical impact have not been presented as established outcomes without supporting empirical evidence.

All external research materials, publications, and other sources used in the final thesis should be appropriately acknowledged and cited according to the referencing style required by the institution.

Use of Artificial Intelligence Tools

Artificial intelligence tools, including ChatGPT, were used as supplementary assistance during the preparation of this research and its documentation. Their use supported activities such as organizing thesis sections, improving academic writing, clarifying technical concepts, refining explanations of the system architecture, and structuring descriptions of implementation and testing results.

AI assistance was also used to help present technical information concerning the XGBoost forecasting model, SHAP-based explainability, LLM-generated explanations, FastAPI integration, and software testing in a clearer academic format.

AI-generated content was treated as drafting and editing assistance rather than as an independent source of verified research evidence. Technical descriptions and evaluation claims were intended to be checked against the implemented system, available test results, and relevant academic literature.

Human Responsibility and Verification

Responsibility for the accuracy, originality, integrity, and final submission of the thesis remains with the researcher. AI-assisted material should be reviewed to ensure that it accurately reflects the implemented component and the evidence obtained during evaluation.

The following principles were applied when preparing the thesis:

- Verification of technical claims: Descriptions of implementation and testing should correspond to the actual system and recorded results.
- Accuracy of reported findings: Numerical measurements and test outcomes should be reported using the available evidence without inventing or overstating results.

- Acknowledgement of limitations: Claims concerning predictive effectiveness, explanation quality, and stakeholder usefulness should remain consistent with the evaluation performed.
- Responsible use of sources: External publications and research materials should be cited appropriately, and references should be verified against the original sources.
- Human review: AI-assisted writing should be reviewed and revised to ensure that the final document reflects the researcher's actual work and understanding.
- Compliance with institutional requirements: The use of AI tools should follow the academic integrity and AI-use disclosure policies applicable to the degree

programme.

